MIL-STD-1665
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SUPERSEDING
(See 6.2)

MILITARY STANDARD

TEST EQUIPMENT FOR
THE
STANDARD ELECTRONIC MODULES PROGRAM



FSC MISC

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DEPARTMENT OF DEFENSE

WASHINGTON, D. C. 20301

Test Equipment for the Standard Electronic Modules Program

MIL-STD-1665

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard establishes the capabilities of electronic test equipment needed for testing of Standard Electronic Modules (as specified in MIL-M-28787). It specifies equipment performance parameters and lists representative test equipment.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Standard Electronic Modules Program, General Specification For.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 Not applicable.

4. GENERAL REQUIREMENTS

4.1 Substitution of equivalent equipment. With the concurrence of the procuring activity, equipment other than that listed herein may be used provided that the substitute equipment has capabilities and performance characteristics at least equal to the listed equipment with respect to test data to be taken. Tests shall be performed as required to verify correlation between the substitute and the listed equipment.

4.2 Significance of asterisk. Test equipment marked with an asterisk (*) shall be considered as reference equipment to be used in the event of conflicting data.

4.3 Arrangement of data and requirements. The requirements and pertinent data pertaining to the specific equipment are arranged as follows:

TITLE - - - - - - - - - - - - - - - -	Lists the item number referenced and specified in detail specifications, tests, etc., and the generic name of the equipment.
TYPE OF EQUIPMENT - - - - - - - - -	Generic name of the equipment.
FUNCTION PERFORMED - - - - - - - -	Defines types of tests within the capability of the equipment.
SPECIFICATIONS - - - - - - - - - -	This section lists the performance parameters and specifications required for the equipment.
EQUIPMENT MEETING ALL SPECIFICATIONS - - - - - - - - -	This section lists specific equipment by manufacturer's name or trade name and model number that complies with or surpasses the minimum requirements listed under specifications.

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EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS - - - - - - - - -

This section lists specific test equipment by manufacturer's name or trade name and model number, together with deviations from the specification, and other pertinent information. Except for the specified deviations, the equipment complies with the specification requirements. The term "exceptions" is construed to mean only those deviations which limit the measurement capability of the equipment.

4.4 General purpose test equipment. Miscellaneous test equipment suitable for general bench testing, and for incorporation in factory production testing equipment, is listed and described under item number designations in the remainder of this document.

4.5 Test equipment application criteria. The choice of test equipment to perform a given measurement should be determined on the basis of the following application criteria. The detailed requirements for maintenance and calibration of inspection and test equipment are defined in MIL-M-28787.

4.5.1 Parameter. The parameter to be measured determines the family of test equipment (ac voltmeters, ohmmeters, frequency meters, etc.) which should be investigated for a specific instrument to perform the measurement.

4.5.2 Range. This criteria determines the qualitative range that the instrument must have in order to measure the magnitude of the parameter.

4.5.3 Accuracy. Once the type of instrument and range have been determined, it is necessary to consider the measurement accuracy required. The accuracy required is a function of the specified parameter tolerance and the required ratio of that tolerance to the test equipment accuracy. The ratio of specification tolerance to test equipment accuracy shall be 10:1 for tolerances of 3 percent or greater and at least 4:1 for tolerances of less than 3 percent.

4.5.4 Other selection criteria. After the necessary considerations of test equipment parameter, range and accuracy have been made, the final selection of a particular manufacturer's instrument is determined as a matter of judgment and compromise of the following:

(a) Secondary features of the insturment.

(b) Multiple usage to satisfy the requirements of other item numbers.

(c) Effect of the instrument on the parameter being measured.

(d) Cost.

4.6 Specifying new test equipment. This document is not intended to be a catalog of manufacturer's general purpose test equipment. Therefore, potential users and contributors to this document, when specifying equipment for the testing of an electronic module, should attempt to restrict their choice to the item numbered equipment in this document. If a suitable test equipment item cannot be found in this document, the user should evaluate his requirements and search for a suitable commercial instrument. After finding a suitable instrument the user should make a written request to the SEMP-QAA to have his test equipment requirements incorporated under an item number in this document.

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The user shall justify his choice of equipment by stating his test requirements, and how this choice satisfies those requirements. This written request shall also include the information described in 4.3 (with the exception of the item number) for his specific equipment. The specification section should not be a duplicate of a manufacturer's specification for a unique instrument. The number of ranges, range accuracy, type of readout display, power requirements, exact input, impedance, color, and instrument weight are examples of specifications for a specific make and model of instrument.

5. DETAILED REQUIREMENTS

5.1 Not applicable.

6. NOTES

6.1 Obsolescent test equipment. As new detail module specifications are developed, the following equipment is not to be called out: G02, H02, R01, R04, R05, T02, T03, V01, V02, Y03, Y06, W02, W03, W06, and X01 (HP412).

6.2 Supersession. This standard supersedes NAVORD OD 36698, Test Equipment.

Custodians:
Navy - EC
Army - EL
Air Force - 11

Review activities:
Navy - AS, OS, SH
Army - AT, MI, MU
Air Force - 13, 17, 19, 85
DLA - ES

User activities:
Army - PA
Navy - MC
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

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ITEM A01

AMPLIFIER, POWER

TYPE OF EQUIPMENT	Power amplifier.
FUNCTION PERFORMED	Provides voltage, power amplification, or voltage and power amplification at low output impedance levels over a broad frequency range. Output is transformer isolated.

SPECIFICATIONS

Input	
Level	0.5 to 30 Vrms with gain control.
Gain	0.5 V for 75 W output.
Impedance	250 kΩ.
Output	
Power	75 W continuous, 20 Hz to 20 kHz.
Frequency response	20 Hz to 20 kHz +0, -0.2 dB at 75 W. 18 Hz to 40 kHz +0, -1 dB at 75 W.
Taps for rated output load impedances	4, 8, 16, 67, 150, and 600Ω nominal. All outputs are isolated from ground.
Output voltages (at maximum power)	25, 70.7, 115, and 230 V.
Output generator impedances	Less than 8 percent of rated load impedance for each output tap.
Nonlinearities	
Harmonic distortion	Less than 0.5 percent at 75 W or less, over frequency range of 20 Hz to 20 kHz.
Intermodulation distortion	Less than 0.5 percent for instantaneous peak power output up to 150 w.
Noise and hum level	90 dB, or more below rated output.

EQUIPMENT MEETING ALL SPECIFICIATIONS

Manufacturer	McIntosh Model #MI-75.

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ITEM A03

AMPLIFIER, LOW NOISE

TYPE OF EQUIPMENT	Low noise, wide band, variable gain amplifier.
FUNCTION PERFORMED	Amplifies signals from 1 Hz to 100 kHz.
SPECIFICATIONS	
Gain	Selectable, 20, 40, or 60 dB (reference 1 V) $\pm$ 0.5 dB.
Bandwidth	1 Hz to 100 kHz flat $\pm$ 1 dB.
Distortion	Less than 0.1 percent.
Input impedance	1 MΩ, 30 pF.
Output impedance	Less than 50 Ω.
Maximum output	1 Vrms @ 1 mA.
Noise	
Noise voltage	Less than 10 uVrms referred to input for full bandwidth with input shorted.
Spectral density	5×10^{-9} volts per root Hz and 3×10^{-12} amperes per root Hz, 1 to 100 kHz.
Power	Mercury batteries - approximately 2,000 hours life.
Terminals	BNC.
EQUIPMENT MEETING SPECIFICATIONS	
Manufacturer	Quan-Tech, Inc., Model 206C.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM 801

ANALYZER, DISTORTION

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Distortion analyzer.

FUNCTION PERFORMED - - - - - - - - - - - - - Measures total distortion, harmonics, and ac voltages over a wide range of frequencies.

SPECIFICATIONS

Distortion measurement accuracy.

(a) Harmonic measurement accuracy.

(1) Fundamental input less than 30 Vrms.

Distortion level range	±3 Percent	±6 Percent	±12 Percent
100 to 0.3 % 0.1%	10 Hz - 1 MHz 30 Hz - 300 kHz	10 Hz - 3 MHz 20 Hz - 500 kHz	(not specified) 10 Hz - 1.2 MHz

(2) Fundamental input greater than 30 Vrms.

Distortion level range	±3 Percent	±6 Percent	±12 Percent
100 to 0.3% 0.1%	10 Hz - 300 kHz 30 Hz - 300 kHz	10 Hz - 500 kHz 20 Hz - 500 kHz	10 Hz - 3 MHz 10 Hz - 1.2 MHz

(b) Elimination characteristics.

(1) Fundamental rejection. Greater than 80 dB.

Second harmonic accuracy	Fundamental frequency
Better than ±1 dB	5 to 20 Hz
Better than ±0.6 dB	20 Hz to 20 kHz
Better than -1 dB	20 to 100 kHz
Better than -2 dB	100 to 300 kHz
Better than -3 dB	300 to 600 kHz

(2) Distortion introduced by instrument - - - - - - - - - - 0.03 percent from 5 Hz to 200 kHz.
0.06 percent from 200 to 600 kHz.

NOTE: Meter indication is proportional to the average value of a sine wave.

(c) Frequency calibration accuracy - - within ±2 percent from 10 Hz to 200 kHz.
within +0 percent, -3 percent from 5 to 10 Hz.
within +8 percent, -0 percent from 200 to 600 kHz.

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Input impedance
Distortion mode - - - - - - - - - - - - - - 1 MΩ shunted by less than 60 pF.
Voltmeter mode - - - - - - - - - - - - - - 1 MΩ shunted by 30 pF (1 to 300 Vrms) 1 MΩ shunted by 60 pF (300 μVrms to 0.3 Vrms).

Input level for distortion measurements - - - - - - - - - - - - - - 0.3 Vrms for 100 percent set level 0.245 Vrms for 0 dB set level (up to 300 Vrms may be attenuated to set level reference).

DC isolation - - - - - - - - - - - - - - - Signal return may be ±400 Vdc from external chassis.

Voltmeter range - - - - - - - - - - - - - - 300 uV to 300 Vrms full scale; 13 ranges; 100 dB per range.

Voltmeter accuracy - - - - - - - - - - -

Range (full scale)	±2 Percent	±5 Percent
300 μV	30 Hz to 300 kHz	20 Hz to 500 kHz
1 mV to 30 V	10 Hz to 1 MHz	5 Hz to 3 MHz
100 V to 300 V	10 Hz to 300 kHz	5 Hz to 500 kHz

Noise measurements - - - - - - - - - - - Voltmeter residual noise on the 300 μV range is: less than 25 μVrms when terminated in 600Ω.
less than 30 μVrms when terminated with a 100 kΩ shielded resistor.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard
Model *331A
Model 332A
Model 333A
Model 334A

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

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ITEM 802

ANALYZER, WAVE

TYPE OF EQUIPMENT	Wave analyzer.
FUNCTION PERFORMED	A tunable ac voltmeter of high selectivity and high sensitivity.

SPECIFICATIONS

Voltage range	30 μV to 300 V full scale, 15 ranges minimum from 30 μV to 300 V in a 3, 1 sequence.
Voltage resolution	3.33 percent of full scale on 3 sequence and 2 percent of full scale on 1 sequence.
Frequency range	20 Hz to 50 kHz.
Frequency resolution	Linear graduation 1 division per 10 Hz.
Frequency accuracy	± (1 percent + 5 Hz) of dial setting.
Voltage accuracy	±5 percent of full scale.
Residual modulation and hum voltage	Greater than -75 dB.
Intermediate frequency rejection	At least -75 dB.
Input impedance	Determined by setting of input attenuator: 100,000Ω shunted by less than 100 pF on four most sensitive ranges, 1 MΩ shunted by less than 20 pF on remaining ranges.
Selectivity	±1 Hz bandwidth less than 1 percent down; ±3-1/2 Hz bandwidth at least 3 dB down; ±25 Hz bandwidth at least 50 dB down; ±70 Hz bandwidth at least 80 dB down; beyond ±70 Hz bandwidth at least 80 dB down.
Restored frequency output	1 volt across 600Ω at output terminals for full scale meter deflection; output voltage proportional to meter reading. Output level control provided. Frequency response ±2 percent, 20 Hz to 50 kHz. Output impedance approximately 600Ω.
Oscillator output	1 volt across 600Ω at output terminals (mode selector in B.F.O.). Output level control provided. Frequency response ±2 percent, 20 Hz to 50 kHz. Output impedance 600Ω.
Recorder output	1 mAdc into 1,500Ω of less at full scale meter indication; for grounded or ungrounded recorders.

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ITEM 803

ANALYZER, SPECTRUM

TYPE OF EQUIPMENT - - - - - - - - - - - - - - - Spectrum analyzer.

FUNCTION PERFORMED - - - - - - - - - - - - - - Frequency selective voltage measurements. CRT display of frequency scan.

SPECIFICATIONS

	Log	Linear
Range		
Linear - - - - - - - - - - - - - - - - - -	100 nV to 20 V full scale.	
Log -	-70 to ± 30 dBV.	
Amplitude accuracy		
Frequency response		
20 Hz to 20 kHz - - - - - - - - - - - - -	± 0.3 dB	±3 percent
5 Hz to 50 kHz - - - - - - - - - - - - - -	± 0.5 dB	±5 percent
Switching between bandwidths (25°C)		
3 Hz to 300 Hz - - - - - - - - - - - - - -	± 0.5 dB	±5 percent
1 Hz to 300 Hz - - - - - - - - - - - - - -	± 1 dB	±10 percent
Amplitude display - - - - - - - - - - - - -	± 2 dB	±2 percent.
Input attenuator - - - - - - - - - - - - - -	± 0.3 dB	±3 percent.
Amplitude reference level (IF attenuator)		
Most sensitive range - - - - - - - - - - -	± 1 dB	±10 percent
All other ranges - - - - - - - - - - - - -	±1 dB	±3 percent
Dynamic range - - - - - - - - - - - - - - - -	80 dB minimum.	
Noise sidebands - - - - - - - - - - - - - - -	Greater than 70 dB below. CW signal, 10 Hz away from signal with 1 Hz bandwidth.	
IF feedthrough		
Input > 10 V - - - - - - - - - - - - - - - -	-60 dB.	
Input < 10 V - - - - - - - - - - - - - - - -	-70 dB.	
Spurious responses - - - - - - - - - - - - -	>80 dB below input reference level.	
Zero beat response - - - - - - - - - - - - -	>30 dB below input reference level.	
Calibrator - - - - - - - - - - - - - - - - -	± 1.5 percent, at full scale, 10 kHz.	
Sweep characteristics		
Scan width - - - - - - - - - - - - - - - - -	50 Hz to 50 kHz.	
Log sweep - - - - - - - - - - - - - - - - - -	20 Hz to 43 kHz ±20 percent.	
Sweep times (linear scan) - - - - - - - - -	1 to 2,000 seconds.	
Accuracy - - - - - - - - - - - - - - - - -	±5 percent.	
Linearity - - - - - - - - - - - - - - - - -	±1 percent.	
Sweep time (log) - - - - - - - - - - - - - -	5 s ±20 percent (not adjustable).	
Sweep linearity - - - - - - - - - - - - - - -	±1 percent.	
Adaptive sweep - - - - - - - - - - - - - - -	When in adaptive sweep below the threshold level, the scan speed is 20 to 25 times faster. Threshold is adjustable to cover 0 to 60 percent of screen. Signals greater than about 6 dB above threshold are detected and swept slowly.	
Input characteristics		
Impedance - - - - - - - - - - - - - - - - - -	1 MΩ, 30 pF.	
Maximum input level - - - - - - - - - - - - -	100 Vrms, ±100 Vdc.	

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B03 - Continued

Output characteristics (tracking generator output)	
Range	0 to 2 Vrms.
Frequency response, 5 Hz to 50 kHz	±3 percent.
Frequency accuracy (relative to center of filter)	±2.5 Hz.
Impedance	600Ω.
Total harmonic and spurious content	40 dB below 1 V signal level.
Frequency characteristics	
Range	5 Hz to 50 kHz.
Frequency dial accuracy	±100 Hz, 20° to 30°C. ±300 Hz, 0° to 55°C.
Display accuracy	Frequency error between any two points is less than ±2 percent of their indicated separation.
Typical stability	±10 Hz/hour after 1 hour; ±5 Hz/°C.
Frequency dial resolution	20 Hz on frequency dial.
Bandwidths	1, 3, 10, 30, and 100 Hz with a shape factor of 10. 300 Hz with a shape factor of 8.
Operating temperature range	0° to 55°C.
Power requirements	100, 120, 220, or 240 V $^{+5\ \text{percent}}_{-10\ \text{percent}}$; 48 to 440 Hz; 35 W maximum.
Options	Battery supply. Rack mount. Floating input.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Hewlett Packard Model 3580A Spectrum analyzer.

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ITEM C01

BRIDGE, CAPACITANCE

TYPE OF EQUIPMENT	Capacitance bridge.
FUNCTION PERFORMED	Measures capacitance, dissipation factor and quality factor of two terminal networks from 20 Hz to 20 kHz.
SPECIFICATIONS	
Capacitance	
Range	0.1 pF to 1,200 uF.
Accuracy	1 percent of reading.
Dissipation factor	
Range	0.0001 to 10,000.
Accuracy	10 percent of reading.
Quality factor	
Range	0.0001 to 10,000.
Accuracy	10 percent of reading.
Frequency range	20 Hz to 20 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH INDICATED EXCEPTIONS AS NOTED:	
Manufacturer	Electro Scientific Industries Model 290A. Impedence bridge used in conjunction with 861A ac generator-detector and 840 (or 803) dc generator-detector (see L01).
Exceptions	Capacitance accuracy - Worst accuracy is 0.2 percent of reading + 1 dial div. +1.0 percent x D x F_{kHz}. Below C = 1 pF, accuracy is worse than 1 percent of reading. Special features - Measures both parallel and series capacitance. Also satisfies requirements of L01.
Manufacturer	General Radio Co., Model 1608-A. Used in conjunction with an ac generator-detector, 1201C and 1232A.
Exceptions	Capacitance range = 0.05 pF to 1,100 uF. Dissipation factor range = 0.0005 to 1 for series C and 0.02 to 2 for parallel C.

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ITEM C02

BRIDGE, CAPACITANCE, RF

TYPE OF EQUIPMENT	Capacitance bridge.
FUNCTION PERFORMED	Measures capacitance of 2-terminal networks at radio frequencies.
SPECIFICATIONS	
Capacitance	
Range	0 to 120 pF.
Accuracy	2 percent of reading.
Frequency range	0.5 to 250 MHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATION WITH EXCEPTIONS AS NOTED	
Manufacturer	Boonton Radio Corp., 250A RX meter (see L03) H.P. Model 250B RX meter.
Exceptions	Accuracy - $\pm 0.5\ (1 + F^2C \times 10^{-5})$ percent ± 0.15 pF with F in MHz and C in pF.

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ITEM C03

BRIDGE, CAPACITANCE

TYPE OF EQUIPMENT	Capacitance bridge.
FUNCTION PERFORMED	Measures capacitance to 11.1 uF and dissipation factor (DF) to 10 percent from 100 Hz to 10 kHz.
SPECIFICATIONS	
Capacitance	
Range	10^{-5} pF to 11.1 uF.
Accuracy	±1 percent of capacitor value.
Dissipation factor (DF):	
Range	Up to 10 percent.
Accuracy	±1 percent of DF reading.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	General Radio Type 1620-A capacitance measuring assembly.
Exceptions	Available signal frequencies over the specified range are 100, 120, 200, 400, 500, 1,000, 2,000, 5,000, and 10,000 Hz.

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ITEM C04

BRIDGE, CAPACITANCE

TYPE OF EQUIPMENT	Capacitance bridge.
FUNCTION PERFORMED	Measures small values of capacitance.
SPECIFICATIONS	
Capacitance	
Range	0 to 300 pF.
Accuracy	±3 percent of reading.
Frequency range	0 to 150 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Tektronix Type 130 L-C meter.
Exceptions	Accuracy - ±3 percent of full scale. Frequency range - 125 to 140 kHz.

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ITEM C05

METER, CAPACITANCE

TYPE OF EQUIPMENT	Capacitance meter.
FUNCTION PERFORMED	Measures small values of capacitance.
SPECIFICATIONS	
Capacitance	
Range	0 to 1,000 pf.
Accuracy	±3 percent of reading.
External D.C. bias	Up to ±200 V at 250 mA.
Frequency range	1 MHz.
Signal level	15 mV.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Boonton Electronics Corporation capacitance/inductance meter Models 71A and 71AR.

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ITEM D01

OSCILLOSCOPE, HIGH FREQUENCY

TYPE OF EQUIPMENT	High-frequency oscilloscope.
FUNCTION PERFORMED	Analog display of electrical signals (dual trace).

SPECIFICATIONS

General information	
Cathode-ray tube	The cathode-ray tube shall present a signal within the accuracy of the horizontal and vertical signal circuits over the full area of 10 cm horizontal by 4 cm vertical. (6 cm vertical is preferred, but not required.) A graticule, either external or internal, shall be supplied for measurements.
Positioning of trace	Provision for positioning the trace or spot and determining its location if off screen shall be incorporated.
Controls	CRT spot size, shape, intensity, graticule brightness, and power on/off.
Calibrator	1 kHz (nominal) square wave from 0.2 mV to 100 V in 1, 2, 5 steps/decade for calibration of vertical circuits and probe alinement.
Horizontal circuits	
Internal sweep	Two sweep (time base) circuits shall be provided.
Time base 1	(a) 100 ns/cm to 5s/cm in 1, 2, 5, sequence, steps/decade (accuracy ±3 percent). (b) A magnifier of at least 5X shall be supplied with sweep accuracy degraded to not more than ±5 percent, for all calibrated fixed step sweep speeds. (c) A noncalibrated vernier shall be capable of slowing sweep from 1:1 to greater than 2:1.
Time base 2	2 us/cm to 1s/cm in 1, 2, 5 sequence, steps/decade.
External	Direct drive of horizontal circuits. (a) Frequency response dc to 225 kHz. (b) Minimum sensitivity 200 mV/cm. (c) Input RC: 1 MΩ shunted by 60 pF maximum.
Triggers	(a) Modes (1) AC (2) AC with low frequency reject (3) DC (b) Source/minimum sensitivity (1) Internal/1 cm display amplitude (2) External/1 volt (3) Line (c) Polarity (1) Positive (2) Negative

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D01 - Continued

Horizontal modes (may be mutually exclusive)- - - - - - - - - - - - - - - -	(a) External (b) Single sweep (1) Manual (time base 1 rate) (2) Manual triggered (time base 1 rate) (c) Time base 1 (d) Time base 2 (e) Time base 1 delayed by (K) x (time base 2) where 0 < K < 1. Jitter: 1/20,000 of time base 2 or less. Delayed sweep presentation may be optionally stabilized.
Horizontal outputs -- - - - - - - - - - -	(a) Time base 1 ramp (b) Time base 1 gate (c) Time base 2 gate and delayed trigger are desirable but optional.
Vertical circuits	
Vertical signal delay - - - - - - - - - -	The vertical signal shall be delayed so that no part of the signal is lost due to start-up time of properly adjusted horizontal sweep circuits.
Vertical sensitivity - - - - - - - - - - -	200 V/CM to 20.0 V/CM
Number of traces - - - - - - - - - - - -	2.
Rise time - - - - - - - - - - - - - - - -	20 ns maximum.
Frequency response - - - - - - - - - - -	DC to 20 MHz.
Polarity reversal - - - - - - - - - - - -	Each trace independent.
Sensitivity - - - - - - - - - - - - - - -	50 mV/cm to 200 V/cm in 1, 2, 5 sequence, 3 steps/decade.
Input impedance - - - - - - - - - - - - -	1 MΩ, shunted by 25 pF, maximum.
Modes - - - - - - - - - - - - - - - - - -	(a) Single trace (either channel) (b) Dual trace (both simultaneously or alternately chopped) (c) Sum (optional)

EQUIPMENT MEETING ALL SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - -	Tektronix, Inc. Any one of the following models: *(1) 545A (out of production, January 1967) *(2) 545B (3) 546 (4) 547 Used in conjunction with one of the following plug-in units: *(1) C-A (2) M (3) 1A2 (4) 1A1

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - -	Hewlett-Packard 140A Used in conjunction with the following plug-in units: 1421A and 1420A
Exceptions - - - - - - - - - - - - - - - -	Minimum sweep speed is 1s/cm calibrated
Manufacturer - - - - - - - - - - - - - - -	Fairchild Any one of the following models: (1) 765H (2) 766H (3) 767H

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D01 - Continued

	Used in conjunction with the following plug-in units and delay line (part No. 7001):
	(1) 74-13A and
	(2) 76-08
Exceptions	(1) Minimum sweep speed 2s/cm calibrated
	(2) No time base 1 ramp output
Manufacturer	Tektronix, Inc., 535 series
	Used in conjunction with one of the following plug-in units:
	(1) C-A
	(2) M
	(3) 1A2
	(4) 1A1
Exceptions	Reduced frequency response to 15 MHz and corresponding increase in rise time.
Manufacturer	Hewlett-Packard
	1720A
	1722A
Exceptions	(1) Vertical sensitivity: 10 mV/division to 5 V/division
	(2) Sweep speeds: normal, 10 ns/division to 0.5 s/division; delayed, 10 ns/division to 20 ms/division
	(3) Inversion: One channel only
	(4) Calibrator: 1 kHz ±10% square wave at 3 V_{p-p} ±1%.
Manufacturer	Tektronix 475
Exceptions	(1) Vertical sensitivity: 2 mV/division to 5 V/division
	(2) Sweep speeds: normal, 10 ns/division to 0.5 s/division; delayed, 5 ns/division to 0.5 s/division
	(3) Inversion: One channel only
	(4) Calibrator: 1 kHz square wave, 300 mV ±1%.

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ITEM D02

OSCILLOSCOPE, HIGH GAIN

TYPE OF EQUIPMENT - - - - - - - - - - - - - - High-gain oscilloscope.

FUNCTION PERFORMED - - - - - - - - - - - - - Analog display of electrical signals.

SPECIFICATIONS:

General information

Cathode-ray tube - - - - - - - - - - - - The cathode-ray tube shall present a signal within the accuracy of the horizontal and vertical signal circuits over the full area of 10 cm horizontal by 6 cm vertical. (6 cm vertical is preferred, but not required.) A graticule, either external or internal, shall be supplied for measurements.

Positioning of trace - - - - - - - - - - Provision for positioning the trace or spot and determining its location if off screen shall be incorporated.

Controls - - - - - - - - - - - - - - - - CRT spot size, shape, intensity, graticule brightness, and power on/off.

Calibrator - - - - - - - - - - - - - - - - 1 kHz (nominal) square wave from 0.2 mV to 100 V in 1, 2, 5 steps/decade for calibration of vertical circuits and probe alinement.

Horizontal circuits

Internal sweep - - - - - - - - - - - - - Two sweep (time base) circuits shall be provided.

Time base 1 - - - - - - - - - - - - - - - (a) 100 ns/cm to 5s/cm in 1, 2, 5 sequence, steps/decade (accuracy ±3 percent)
(b) A magnifier of at least 5X shall be supplied with sweep accuracy degraded to not more than ±5 percent, for all calibrated fixed step sweep speeds.
(c) A noncalibrated vernier shall be capable of slowing sweep from 1:1 to greater than 2:1.

Time base 2 - - - - - - - - - - - - - - - 2 µs/cm to 1s/cm in 1, 2, 5 sequence, steps/decade.

External - - - - - - - - - - - - - - - - Direct drive of horizontal circuits
(a) Frequency response dc to 225 kHz.
(b) Minimum sensitivity 200 mV/cm.
(c) Input RC: 1 MΩ shunted by 60 pF maximum.

Triggers - - - - - - - - - - - - - - - - (a) Modes:
(1) AC
(2) AC with low frequency reject
(3) DC
(b) Source/minimum sensitivity
(1) Internal/1 cm display amplitude
(2) External/1 volt
(3) Line

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	(c) Polarity (1) Positive (2) Negative
Horizontal modes (may be mutually exclusive)	(a) External (b) Single sweep (1) Manual (time base 1 rate) (2) Manual triggered (time base 1 rate) (c) Time base 1 (d) Time base 2 (e) Time base 1 delayed by (K) x (time base 2) where 0 < K < 1. Jitter: 1/20,000 of time base 2 or less. Delayed sweep presentation may be optionally stabilized.
Horizontal outputs	(a) Time base 1 ramp (b) Time base 1 gate (c) Time base 2 gate and delayed trigger are desirable but optional.
Vertical circuits	
Vertical signal delay	The vertical signal shall be delayed so that no part of the signal is lost due to start-up time of properly adjusted horizontal sweep circuits.
Sensitivity	1 mV/cm at 0.30 MHz with a rise time of 1 us to 50 mV/cm at 2 MHz with a rise time of 200 ns in 3 steps/decade 1, 2, 5 sequence.
Attenuation	1 to 1,000 in decade steps.
Polarity	Normal or inverted.
Differential	The differential input provides common mode rejection to enable accurate measurement of difference between two simultaneous signals. (This is not a requirement. However, it is a useful refinement.)
Output	Internal vertical signal available at front panel.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Tektronix, Inc.
	Any one of the following models: (1) 535A (or 535) (2) 545A (out of production, January 1967) *(3) 545B (4) 546 (5) 547
	Used in conjunction with one of the following plug-in units: (1) D or (2) W (requires D6023 probe for some tests)

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Hewlett-Packard
	140A with a 1401A plug-in unit.
Exceptions	Slowest sweep rate, 1s/cm.

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ITEM D03

OSCILLOSCOPE, HIGH FREQUENCY

TYPE OF EQUIPMENT	High-frequency oscilloscope.
FUNCTION PERFORMED	Analog display of electrical signals.
SPECIFICATIONS	
General information	
Cathode-ray tube	The cathode-ray tube shall present a signal within the accuracy of the horizontal and vertical signal circuits over the full area of 10 cm horizontal by 4 cm vertical. (6 cm vertical is preferred, but not required.) A graticule, either external or internal, shall be supplied for measurements.
Positioning of trace	Provision for positioning the trace or spot and determining its location if off screen shall be incorporated.
Controls	CRT spot size, shape, intensity, graticule brightness, and power on/off.
Calibrator	1 kHz (nominal) square wave from 0.2 mV to 100 V in 1, 2, 5 steps/decade for calibration of vertical circuits and probe alinement.
Horizontal circuits	
Internal sweep	Two sweep circuits shall be provided.
Time base 1	(a) 100 ns/cm to 5s/cm in 1, 2, 5 sequence, steps/ decade (accuracy ±3 percent). (b) A magnifier of at least 5X shall be supplied with sweep accuracy degraded to not more than ±5 percent, for all calibrated fixed step sweep speeds. (c) A noncalibrated vernier shall be capable of slowing sweep from 1:1 to greater than 2:1.
Time base 2	2 us/cm to 1s/cm in 1, 2, 5 sequence, steps/decade.
External	Direct drive of horizontal circuits. (a) Frequency response dc to 225 kHz. (b) Minimum sensitivity 200 mV/cm. (c) Input RC: 1 MΩ shunted by 60 pF maximum.
Triggers	(a) Modes (1) AC (2) AC with low frequency reject (3) DC (b) Source/minimum sensitivity (1) Internal/1 cm display amplitude (2) External/1 volt (3) Line (c) Polarity (1) Positive (2) Negative

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Horizontal modes (may be mutually exclusive)	(a) External (b) Single sweep (1) Manual (time base 1 rate) (2) Manual triggered (time base 1 rate) (c) Time base 1 (d) Time base 2 (e) Time base 1 delayed by (K) x (time base 2) where 0 < K < 1. Jitter: 1/20,000 of time base 2 or less. Delayed sweep presentation may be optionally stabilized.
Horizontal outputs	(a) Time base 1 ramp (b) Time base 1 gate (c) Time base 2 gate and delayed trigger are desirable but optional.

Vertical circuits

Delay	The vertical signal shall be delayed so that no part is lost due to sweep circuit start-up time.
Slide back voltmeter	
Accuracy	-100 V to +100 V ±0.75 percent of reading with 10 mV resolution.
Rise time	30 ns.
Frequency response	DC to 13 MHz. AC 5 Hz to 13 MHz.
Sensitivity	0.05 to 25V/cm in 1, 2, 5 sequence/decade.
Input impedance	1 MΩ shunted by 25 pF, maximum.
Number of inputs	Two (2) each having ac or dc coupling.
Output	Internal vertical signal available at front panel.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Tektronix, Inc.
	Any one of the following models: (1) 535A (2) 545A *(3) 545B (4) 546 (5) 547
	Used in conjunction with either of the following plug-in units: *(1) Z or (2) W (requires probe P6023 for some tests).

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Fairchild
	Any one of the following models: (1) 765H (2) 766H (3) 767H
	Used in conjunction with plug-in units: (1) 74-13A and (2) 76-07
Exceptions	(1) Minimum sweep speed of 2s/cm (2) No ramp output (3) No external calibration

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ITEM D04

OSCILLOSCOPE, UHF SAMPLING SYSTEM

TYPE OF EQUIPMENT	UHF sampling system.
FUNCTION PERFORMED	Visual analog (CRT) presentation of signals.

SPECIFICATIONS

General information	
Accuracy (excluding any probes)	Worst case for any measurement ±5 percent assuming optimum setting of controls.
CRT display	(a) Edge lighted internal graticule. (b) 10 cm horizontal by 8 cm vertical.
Calibrator (pulse)	(a) Frequency 20 kHz ±0.1 percent. (b) Voltage (1) 0.5 and 5.0 V, ±2 percent for $R_L \geq 100$ kΩ. (2) 50 and 500 mV, ±2 percent for $R_L \geq 50$ ±0.5Ω.
Controls	Spot geometry, scale illumination, and power on/off.
Horizontal circuits	
Time base (sampling)	(a) Sweep rates (equivalent real time): (1) 1 ns/cm to 200 µs/cm in 1, 2, 5 sequence, 3 steps/decade (calibrated to 3 percent). (2) Uncalibrated continuously variable 1 ns/cm to 20 µs/cm. (b) Sweep rates (real time): Automatic switchover from equivalent real time at sweep speed slower than 200 µs/cm or capable of such operation through use of external sampling clock, trigger, and sweep source.
Modes	(a) Trigger: Internal minimum sensitivity, 50 mV signal. External sensitivity 5 mV. (b) Sweep-repetitive, manual local, manual remote, single sweep. (c) Magnified sweep, X1 or X10. (d) Delayed sweep - minimum of 10 cm for all sweep rates from 1 ns/cm through 100 µs/cm. (e) Samples: 10 or 100/cm (nonmagnified)
Output signals	(a) Trigger 1. (b) Ramp (linear and/or staircase).
Vertical circuits (dual channel sampling)	Unless otherwise noted, all remarks apply to each channel.
Sensitivity	(a) 10, 20, 50, 100, 200 mV/cm ±3 percent. (b) 2, 5 mV/cm ±5 percent. (c) Variable: 2 through 200 mV/cm uncalibrated. (d) Maximum input ±2 V p-p referenced to ground. (e) Noise (max) 2 mV p-p (1 mV smoothed). (f) Input impedance (50 ±0.5) ±10Ω. (g) Signal delayed to allow horizontal circuits to start plus 10 ns.

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D04 - Continued

Frequency response	(a) 0.4 ns rise time. (b) DC to 875 MHz.
Modes (channels A, B)	(a) Either channel (A or B). (b) Dual channel (A and B). (c) Sum (±A ±B). (d) XY (A versus B, A vertical).
DC offset voltage (slide back voltmeter)	±1 Vdc with provision for monitoring.
Vertical signal outputs	(a) Channel A. (b) Channel B. (c) ±A ±B when in sum mode.

Probe

Input impedance - 10 MΩ shunted by: (all approximate)

Attenuation	DC	AC	Maximum in volts (1)
X1	4 pF	6 pF	±2 V
X10	2 pF	4 pF	±20 V
X100	1.6 pF	3.6 pF	±100 V

(1) At 260 MHz (dc coupled or p-p ac coupled).

Bandpass	(a) DC coupling dc to 230 MHz (b) AC coupling 16 Hz to 230 MHz
Pulse response	(a) Rise time 1.5 ns maximum. (b) Aberrations, total ±4.0 percent.
Gain	(a) X1 Attenuation adjustable 0.9 to 1.1. (b) X10 and X100 attenuation within ±3.5 percent of X1 attenuation.
Noise	1.5 mV peak-to-peak.
Output impedance	50Ω.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Tektronix, Inc. Model 561 or 561A with 3T4 horizontal module and 3S76 with two P6045 probes.
Item	D05

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Analab 1120 main frame with 701 vertical/horizontal module and two 9009 probes.
Exceptions	(1) Probe capacitance not specified. (2) No provision for real time sampling.
Manufacturer	Tektronix, Inc. 561 or 561A main frame with 3T4 or 3T77A horizontal module and 3S3 vertical module.
Exceptions	Input resistance 100 kΩ; capacitance 2 pF.
Manufacturer	Tektronix, Inc. 661 main frame with 5T1, 5T1A, 5T3 horizontal modules and 4S2 or 4S2A vertical module and two P6045 probes, factory adjusted for 50Ω output.
Exceptions	4 mV noise unsmoothed. 2 mV noise smoothed. 5T1, 5T1A - no provision for real time sampling.

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DO4 - Continued

Manufacturer	Tektronix, Inc. 661 main frame with 5T1, 5T1A, 5T3 horizontal module and 453 vertical modules.
Exceptions	Input resistance: 100 kΩ. Capacitance: 2 pF. Maximum attenuator X10. 5T1, 5T1A, no real time sampling provisions.
Manufacturer	Hewlett-Packard. 140A main frame with 1424A horizontal module and 1411 vertical module with 1432A sampler.
Exceptions	No high impedance probes, no real time sampling provisions.
Manufacturer	Hewlett-Packard. 140A main frame with 1424A horizontal module and 1410A vertical module.
Exceptions	Input resistance: 100 kΩ. Capacitance: 2 pF. No provisions for real time sampling.

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ITEM D05

OSCILLOSCOPE, UHF SAMPLING SYSTEM

TYPE OF EQUIPMENT	UHF sampling system.
FUNCTION PERFORMED	Visual analog (CRT) and digital presentation of voltage and time measurements with go/no-go indication with limited pre-programming capabilities.

SPECIFICATIONS

General information

Absolute accuracy (excluding any probes)	Worst case for any measurement ±5 percent assuming optimum setting of controls.
CRT display	(a) Edge lighted internal graticule. (b) 10 cm horizontal by 8 cm vertical.
Calibrator (pulse)	(a) Frequency 20 kHz ±0.1 percent. (b) Voltage (1) 0.5 and 5.0 V, ±2 percent with R_L = 100 kΩ. (2) 50 and 500 mV, ±2 percent with R_L = 50 ±0.5Ω load.
Controls	Spot geometry, scale illumination, and power on/off.

Horizontal circuits

Time base (sampling)	(a) Sweep rates (equivalent real time): (1) 1 ns/cm to 200 µs/cm in 1, 2, 5 sequence, 3 steps/decade (calibrated to 3 percent). (2) Uncalibrated continuously variable 1 ns/cm to 20 µs/cm. (b) Sweep rated (real time): Real time sampling shall enable reduction of sweep rate to 1s/cm with external sampling clock, trigger, and sweep source.
Modes	(a) Trigger: Internal sensitivity, 50 mV. External sensitivity, 5 mV. (b) Sweep - Repetitive; manual local; manual remote: single sweep. (c) Magnified sweep - X1 or X10 (affects horizontal display but not output to digital display unit). (d) Delayed sweep - Minimum of 10 cm for all sweep rates from 1 ns/cm through 100 µs/cm. (e) Samples: 10 to 100/cm (nonmagnified). (f) Remote programming.
Remote programming	(a) Equivalent time or real time. (b) Sweep rate (equivalent time only). (c) Samples per sweep. (d) Repetitive or single sweep modes. (e) Single display start. (f) Digital unit digits and decimals.

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D05 - Continued

Output signals	(a) Trigger 1. (b) Ramp (linear and/or staircase).
Vertical circuits (dual channel sampling)	Unless otherwise noted, all remarks apply to each channel.
Sensitivity	(a) 10, 20, 50, 100, 200 mV/cm ±3 percent. (b) 2, 5 mV/cm ±5 percent. (c) Variable: 2 through 200 mV/cm uncalibrated. (d) Maximum input ±2 V p-p referenced to ground. (e) Noise (max) 2 mV p-p (1 mV smoothed). (f) Input impedance (50 ±0.5) ±10%. (g) Signal delayed to allow horizontal circuits to start plus 10 ns.
Frequency response	(a) 0.4 ns rise time. (b) DC to 875 MHz.
Modes (channels A, B)	(a) Either channel (A or B). (b) Dual channel (A and B). (c) Sum (±A ±B). (d) XY (A versus B, A vertical).
DC offset voltage (slide back voltmeter)	±1 Vdc at monitor jack.
Vertical signal output jacks	(a) Channel A. (b) Channel B. (c) ±A ±B when in sum mode.
Digital readout	
Time measurement	(a) Preselected (fixed increments). (1) Rise time (10-90 percent). (2) Fall time (10-90 percent). (3) Equivalent time constant. (4) Pulse length. (5) Period. (6) Delay. (b) Preselected adjustable. (1) From any point on a given signal to any other point on the same signal (single channel). (2) From any point on a given signal to any point on another signal (dual channel).
Voltage	(a) Peak-to-peak. (b) Average.
Go/no-go operation	Upper and lower limit set controls for any measurement in time measurement or voltage.
Display	(a) 4 digit readout with proper decimal and units. (b) Go/no-go: 3 lights indicating whether measured value is less than minimum setting, within set range, or greater than maximum setting (low no-go/go/high no-go).
Probe	
Input impedance	10 MΩ shunted by: (all approximate)

Attenuation	DC	AC	Maximum in volts (1)
X1	4 pF	6 pF	±2 V
X10	2 pF	4 pF	±20 V
X100	1.6 pF	3.6 pF	±100 V

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	(1) Ac 260 MHz (dc coupled or p-p ac coupled)
Bandpass	(a) DC coupling dc to 230 MHz
	(b) AC coupling 16 Hz to 230 MHz
Pulse response	(a) Rise time, 1.5 ns maximum
	(b) Aberrations, total ±4.0 percent.
Gain	(a) X1 attenuation adjustable 0.9 to 1.1.
	(b) X10 and X100 attenuation within ±3.5 percent of X1 attenuation.
Noise	1.5 mV peak-to-peak.
Output impedance	50Ω.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Tektronix, Inc. (Only known source.) Includes as a minimum all of the following:

Quantity	Part	Model or type no.
1	Main frame	*567
1	Horizontal circuits	*3T4 (Note 1)
1	Vertical circuits	*3S76 (Note 1)
1	Digital unit	*6R1A (Note 1)
2	Probe/probe power supply	*P6045 (w/supply 010-0205-00) (Note 2)

Note 1: Plug-in modules to main frame.
Note 2: Must be factory compensated for 50 system which is load imposed by 3T4 input.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED - - - - - - - - - - None.

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ITEM D06

OSCILLOSCOPE, HIGH FREQUENCY

TYPE OF EQUIPMENT		Storage oscilloscope.
FUNCTION PERFORMED		Visual analog presentation of signals with capability of storing/retaining visual display.
SPECIFICATIONS		
General information		
CRT display	(a)	Edge lighted graticule.
	(b)	Storage time variable from 0.5 s to 5 s with automatic erase.
	(c)	Maximum viewing time 1 hour without erase.
	(d)	Writing speed 0.5 cm/us to 5 cm/us with enhancement.
	(e)	Erase controlled manually, or automatically at end of sweep, or automatically at end of sweep after variable delay listed in (b).
	(f)	Storage possible on either upper or lower half of screen with nonstorage on other half. Also, possible to operate with entire screen store or nonstore.
Horizontal circuits		Two time base circuits are provided.
Time base 1	(a)	0.1 us/cm to 5s/cm in 24 calibrated steps, accurate within 3 percent.
Time base 2	(a)	2 us/cm to 1s/cm in 18 calibrated steps, accurate within 3 percent.
Magnifier	(a)	X5 magnifier operates over full time base, accurate to 5 percent.
	(b)	Increases fastest time base 1 to 20 ns/cm.
	(c)	Increases fastest time base 2 to 0.4 us/cm.
Delayed sweep	(a)	Delay range 2 us to 1 s, accurate within 1 percent.
	(b)	Delay multiplier extends range from 0.10 to 10.0X.
Operating modes	(a)	Time base 1.
	(b)	Time base 2.
	(c)	Time base 2 intensified by time base 1.
	(d)	Time base 1 delayed by time base 2.
Trigger circuits		
Trigger modes	(a)	Automatic, useful between 50 Hz and 1 MHz.
	(b)	Manual.
Coupling	(a)	AC
	(b)	DC
	(c)	AC low-frequency reject.
Sources	(a)	Internal from vertical amplifier.
	(b)	External.
	(c)	Line.

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D06 - Continued

Time base 1 requirements	(a)	0.2 cm deflection internal or 0.3 V external from dc to 10 MHz increasing to 2 cm deflection internal or 3 V external at 30 MHz.
	(b)	Trigger level requirements increase below 300 Hz with ac coupling, below 200 kHz with ac low-frequency reject.
	(c)	Auto operation requires 0.2 cm deflection internal or 0.3 V external from 300 Hz to 10 kHz increasing to 2 cm/3 V at 1 MHz.
Time base 2 requirements	(a)	0.2 cm deflection internal or 0.3 external from dc to 1 MHz increasing to 1 cm deflection internal or 1.5 V external at 10 MHz.
	(b)	Trigger level requirements increase below 300 Hz with ac low-frequency reject.
	(c)	Auto operation requires 0.2 cm deflection internal or 0.3 V external from 300 Hz to 10 kHz increasing to 2 cm/3 V at 1 MHz.
Vertical circuits		
Bandwidth		DC to 30 MHz maximum with 50 mV/cm deflection.
Risetime		12 ns minimum with 50 mV/cm deflection.
Input impedance		1 MΩ paralleled by 15 pF (for each of the dual inputs).
Deflection factor		5 mV/cm to 20 V/cm in 12 calibrated steps. Accurate within 3 percent.
Operating modes	(a)	Channel 1.
	(b)	Channel 2.
	(c)	Algebraic sum, channel 1 + channel 2.
	(d)	Alternate sweep between channels.
	(e)	Switch sweep between channels at approximately 1 MHz rate.
Inverted operation		Each channel will have an invert capability for reversing polarity.
Coupling	(a)	AC
	(b)	DC
Calibrator	(a)	0.2 μV to 100 V square wave, 18 calibrated steps, 2 percent accurate. 1 kHz ± 25 percent repetition rate. Approximately 1 μs risetime.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer		Tektronix, Inc.
	*(1)	Type 549 used in conjunction with the following plug-in:
	*(1)	1A1
	*(2)	1A7

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EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Hewlett Packard, Incorporated
	(1) 181A used in conjunction with the following plug-ins:
	(1) 1801A option 0001, 1822A
Exceptions	(1) Bandwidth dc - 5 MHz.
	(2) Time base range 0.05 us/cm to 1 s/cm.
	(3) Delay multiplier range 1X to 10X.
Manufacturer	Tektronix 4648 with P6062A probe
Exceptions	(1) Storage on entire screen only
	(2) Sweep speeds: normal, 50 ns/division to 0.5 s/division; delayed, 50 ns/division to 50 ms/division
	(3) Input impedance: 1 MΩ, 20 pF
	(4) Deflection sensitivity: 0.3/division
	(5) Inversion: One channel only
	(6) Calibrator: 1 kHz square wave, 300 mV.

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ITEM E01

TRANSFORMER, ISOLATION

TYPE OF EQUIPMENT	Isolation transformer.
FUNCTION PERFORMED	Used in bridging isolation and calibration applications.
SPECIFICATIONS	
Turns ratio	1:1, center tap.
Frequency range	50 Hz to 10 kHz.
DC resistance	10Ω nominal.
Open circuit characteristics (400 Hz)	
Ratio accuracy	0.005 percent.
Phase error	Less than 0.3 mrad.
Input impedance	Greater than 260 kΩ.
Input voltage	0.35 x F (170 V maximum) (F in Hz).
*Common mode rejection ratio with a common mode voltage of 10 Vrms	105 dB (400 Hz) R_L = 10 kΩ. 95 dB (800 Hz) R_L = 10 kΩ. 90 dB (10 kHz) R_L = 10 kΩ.
Shielding	Single electrostatic box shield.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Gertsch Model *ST 200A
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Gertsch Model ST 248A
Exceptions	*(1) Common mode rejection ratio with a common mode voltage of 10 Vrms: 80 dB (400 Hz) R_L = 10 kΩ. 80 dB (800 Hz) R_L = 10 kΩ. 55 dB (10 kHz) R_L = 10 kΩ.

* The equipment specified meets this requirement although the manufacturer's specification does not include this parameter. Verified by measurement.

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ITEM E02

TRANSFORMER, ISOLATION

TYPE OF EQUIPMENT	Isolation transformer.
FUNCTION PERFORMED	Used for precision bridging and isolation.
SPECIFICATIONS	
Turns ratio	4:1.
Frequency	*20 Hz to 5 kHz, usable up to 10 kHz.
DC resistance	3 kΩ (high side), 120Ω (low side).
Input	
Impedance	Greater than 15 kΩ at 400 Hz, secondary open circuit.
Voltage	Up to 115 Vrms at 60 Hz into high side when load across low side is greater than 100Ω.
*Phase error	
*Ratio accuracy	
Shielding	Separate primary and secondary electrostatic box shielding.
*Common mode rejection ratio with a common mode voltage of 200 Vrms	65 dB (400 Hz) R_L = 100 kΩ. 60 dB (800 Hz) R_L = 100 kΩ. 55 dB (10 kHz) R_L = 100 kΩ.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Gertsch Model ST-100.

* The transformer specified meets the requirements although the manufacturer's specification does not include this parameter. Verified by measurement.

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ITEM E03

TRANSFORMER, ISOLATION

TYPE OF EQUIPMENT	Medium frequency isolation transformer.
FUNCTION PERFORMED	Used for isolation of low voltage, medium frequency sinusoids.
SPECIFICATIONS	
Frequency range	100 kHz to 1 MHz.
Flatness	±1 dB over frequency range.
Voltage range	0 to 2 volts peak-to-peak.
Turns ratio	1:1:1 ±1 percent.
Common mode rejection	46 dB at 500 kHz (minimum)
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Caledonia W-7233 pulse transformer.

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ITEM E04

TRANSFORMER, SCOTT "T"

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Scott "T" transformer.

FUNCTION PERFORMED - - - - - - - - - - - - - Provides conversion from resolver four line input to synchro three line input within 1.5 arc minutes.

SPECIFICATIONS

Frequency - - - - - - - - - - - - - - - - - 400 Hz ±5 percent.

Primary voltage - - - - - - - - - - - - - - 7 Vrms ±15 percent.

Primary dc current - - - - - - - - - - - - 2 mA maximum.

Output - - - - - - - - - - - - - - - - - - (a) Secondary voltage (terminals 1-2): 90 Vrms L-L ±3 percent measured at 400 Hz with a voltage (approximately 7.5 Vrms) on a 6-7 which causes 2.8 Vrms to appear across terminals 5-4 and a 6 kΩ resistive load on terminals 1-2, 1-13, and 2-13.

(b) Voltage 3-12: 1.134 Vrms ±0.1 percent, measured with terminals 12-7 shorted and a 7 Vrms signal on terminals 6 to 12-7. The voltage on terminal 3 to be 16.14 percent of the voltage on terminal 6, with all other windings open circuit.

(c) Voltage ratio 6-7 to 4-5 = 2.5:1 ±0.01 percent, measured with terminals 7-4 shorted and a 7 Vrms signal on terminal 6 to 7-4. The voltage on terminal 5 to be 40 percent ±0.01 percent of the voltage on terminal 1, with all other windings open circuit.

Null voltage - - - - - - - - - - - - - - - - 80 mV maximum.

DC resistance - - - - - - - - - - - - - - - DC resistance at +25°C
1-2 = 225Ω ±10 percent
2-13 = 225Ω ±10 percent
1-13 = 225Ω ±10 percent
3-12 = 1.7Ω ±10 percent
11-12 = 1.7Ω ±10 percent
6-7 = 1.7Ω ±10 percent PRI
5-4 = 4.2Ω ±10 percent FB
8-7 = 1.7Ω ±10 percent PRI
9-10 = 4.2Ω ±10 percent FB

Transformation ratio - - - - - - - - - - - ±1.5 arc minutes maximum under load conditions.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - - Perkin Elmer Scott "T" Transformer -03-1276 (Note: IBM Model No. 6096798).

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ITEM E05

TRANSFORMER, ISOLATION

TYPE OF EQUIPMENT	Isolation transformer.
FUNCTION PERFORMED	Used for precision bridging and isolation.

SPECIFICATIONS

Turns ratio	4:1
Frequency	*20 Hz to 5 kHz, usable up to 10 kHz.
DC resistance	3 kΩ (high side), 120 Ω (low side).
Input:	
Impedance	Greater than 15 kΩ at 400 Hz, secondary open circuit.
Voltage	Up to 115 Vrms at 60 Hz into high side when load across low side is greater than 100 Ω.
* Phase error	
* Ratio accuracy	
Shielding	Separate primary and secondary electrostatic box shielding.
* Common mode rejection ratio	65 dB (400 Hz) R_L = 100 kΩ.
with a common mode voltage of 200 Vrms.	60 dB (800 Hz) R_L = 100 kΩ.
	55 dB (10 kHz) R_L = 100 kΩ.
Terminals	Shielded mating connector screws on for a continuous shield between connecting wire and transformer.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Gertsch Model ST-100A.

* The transformer specified meets the requirements although the manufacturer's specification does not include this parameter. Verified by measurement.

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ITEM EL01

LOAD, ELECTRONIC

TYPE OF EQUIPMENT	Electronic load.
FUNCTION PERFORMED	Provides accurate simulation of electrical loads.

SPECIFICATIONS

Load voltage	0 to 50 V.
Load current	0 to 60 A.
Power dissipation	1 to 1000 W.
Overload rating	10 percent.
Self-protection	Overvoltage: 60 V Overcurrent: 65 A Over power: 1300 W
Resistance range	0 to 1 A/V, 0 to 10 A/V (load current proportional to load voltage).
Constant current ranges	0 to 10 A, 0 to 50 A.
Pulse load	0 to 50 A.
External modulation	0 to 6 V.
Pulse modulation	Square wave, frequency adjustable, 500 to 5000 Hz, 0 to full load plus selected steady state load.
Rise time	Less than 50 us.
Input power	105 to 125 V, 47 to 63 Hz.
Voltmeter ranges	6 V, 18 V, 60 V full scale.
Ammeter ranges	6 A, 18 A, 60 A full scale.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Dynaload Model DLP 50-60-1000.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Manufacturer	Dynaload Model DLP 50-150-3000.
Exceptions	Load current: 180 A Power dissipation: 3000 W Resistance ranges: 0 to 5 A/V, 0 to 30 A/V Constant current ranges: 0 to 30 A, 0 to 150 A Pulse load: 0 to 150 A Overcurrent: <180 A Overpower: <3500 W Ammeter ranges: 18 A, 60 A, 180 A full scale.

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ITEM F01

TRANSFORMER, RATIO

TYPE OF EQUIPMENT	Ratio transformer.
FUNCTION PERFORMED	Provides precision ac voltage division.
SPECIFICATION	
Ratio range	0.000000 to 1.11110.
Terminal linearity	0.001 percent.
Accuracy	(a) 50 Hz to 3 kHz: $\pm 0.001\% + \frac{0.0006\%}{\text{Ratio}}$
	(b) 3 to 10 kHz: $\pm 0.01\% + \frac{0.006\%}{\text{Ratio}}$
Input	
Approximate inductance	150 H (F = 100 Hz).
Approximate impedance	250 kΩ (F = 400 Hz).
Maximum voltage	0.35 x F (F in Hz), 350 V maximum.
Resolution	Continuous.
Readout	6 digits.
Maximum effective series impedance	R series, 14Ω; L series, 75 μH.
Phase shift	Less than 1 minute (F = 400 Hz).
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Gertsch Model *CRT-4F.
Item	F02.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	North Atlantic Model RB-503 Model RB-521
Exceptions	Accuracy: 3 to 10 kHz is not specified, but termed useful.
Manufacturer	Electro Scientific Industries (ESI) Model DT 1145
Exceptions	Accuracy: $\pm(1 + 0.1\ F^2_{\text{kHz}})$ x dial divisions, where 1 dial division equals 0.001 percent of full scale.

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ITEM F02

TRANSFORMER, RATIO

TYPE OF EQUIPMENT	Ratio transformer.
FUNCTION PERFORMED	Provides precision ac voltage division.
SPECIFICATIONS	
Range	
Ratio	-0.111110 to +1.111110.
Frequency	50 Hz to 10 kHz.
Accuracy	$\pm(0.0001 + \frac{0.000025}{\text{Ratio}})\%$ at 400 Hz.
Input	
Impedance	>200 kΩ at 400 Hz.
Voltage (maximum)	0.7 x F volts (F in Hz); not to exceed 350 Vrms.
Nominal output resistance (maximum)	8.5Ω.
Phase shift	Less than 1 minute at 400 Hz.
Resolution	Continuous.
Readout	5 digits plus 1 turn potentiometer.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	North Atlantic Model *RB-504.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Gertsch Model RT-19, Model 1011R
Exceptions	1. Maximum input volts = (0.35 x F) Vrms. (F in Hz); not to exceed 350 Vrms. 2. Accuracy: (a) $\pm(0.0001 + \frac{0.000025}{\text{Ratio}})\%$ for 50 Hz to 1 kHz. (b) $\pm(0.0001 + \frac{0.000025}{\text{Ratio}})$ x (F^2kHz)% above 1 kHz.
Manufacturer	Electro Scientific Industries (ESI) Model DT72A
Exceptions	Accuracy: Meets specifications from 50 Hz to 1 kHz; not specified above 1 kHz.

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ITEM F03

DECADE, SYNCHRO/RESOLVER STANDARD

TYPE OF EQUIPMENT	Resolver/synchro standard.
FUNCTION PERFORMED	Provides direct read out of angular positions.
SPECIFICATIONS	
Frequency	400 to 800 Hz.
Angular range	0° to 360° continuous.
Angular accuracy	±2 seconds of arc.
Angular increments	0.0001 degrees.
Excitation voltage	115 or 26 Vac.
Maximum Stator output voltage	115, 90, 26, and 11.8 Vrms.
Maximum effective output impedance unbalance (at 400 Hz)	(1) 115 and 90 V .... 0.05 + J.005Ω. (2) 26 and 11.8 V .... 0.01 + J.001Ω.
Effective variation in transformation ratio from nominal as a function of angular position	±0.2 percent.
Readout	7 digits.
Output	4 ranges: (1) 11.8 Vac. (2) 26 Vac. (3) 50 Vac. (4) 115 Vac.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Gertsch, Decade Synchro/Resolver Standard DSRS-SDR.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

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ITEM F04

RESOLVER/SYNCHRO BRIDGE

TYPE OF EQUIPMENT	Resolver/synchro bridge.
FUNCTION PERFORMED	Provides precision measurements of the angular shaft position of resolvers/synchros.
SPECIFICATIONS	
Frequency range	(a) (within accuracy) 400 Hz. (b) (useful) 400 to 1,600 Hz.
Accuracy (unloaded)	±2 arc seconds or ±0.00055 degrees.
Input impedance (at 400 Hz)	Synchro 500 kΩ. Resolver 750 kΩ.
Resolution	0.0001 degrees.
Voltage (L-L)	Nominal 11.8 to 115 Vrms and a maximum of 150 Vrms or 0.8 F x V (F in Hz), whichever is less.
Null voltage gradient	uV/s = 0.55 x (L-L voltage input). uV/0.001° = 2.0 x (L-L voltage input).
Input mode	Switch between synchro or resolver operation. (Note: This switch also switches real terminals).
Readout	7 digits with 5 knobs.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	North Atlantic Resolver/Synchro Bridge Model 540/10.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM F0S

, TRANSFORMER, RATIO

TYPE OF EQUIPMENT	Ratio transformer.
FUNCTIONS PERFORMED	Provides precision ac voltage division.
SPECIFICATIONS	
Range	
Ratio	0.000000 to 1.111110.
Frequency	5 kHz to 20 kHz.
Terminal linearity	10 ppm at 10 kHz.
Input	
Impedance	50 kΩ (min) at 10 kHz.
Maximum voltage	0.05 x f (f in Hz); not to exceed 350 Vrms.
Resolution	1 ppm.
Output impedance	2 + j1 Ω (max) at 10 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Singer Model RT-25.

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ITEM G01

GENERATOR, SINE WAVE

TYPE OF EQUIPMENT	Sine wave generator.
FUNCTION PERFORMED	Provides accurate variable frequency, sine wave and sync pulses.
SPECIFICATIONS	
Frequency	
Range	0.0015 Hz to 1 MHz.
Voltage - controlled generator (VCG)	4.75 V input for 20:1 frequency ratio. Input impedance is 10 kΩ.
Electronic frequency vernier	1 turn for 1 minor dial division.
VCG bandwidth	100 kHz.
Accuracy (frequency)	
Dial	±1 percent of full scale 0.0015 Hz to 100 kHz. ±2 percent of full scale 100 kHz to 1 MHz.
VCG linearity (0.0015 Hz to 100 kHz)	±0.1 percent frequency versus input voltage (frequency error 0.1 percent of total deviation - best straight line method) from 10 to 100 percent of maximum dial frequency.
Stability (frequency)	
Room temperature	Drift less than ±0.05 percent of setting for 10 minutes. Drift less than ±0.25 percent of setting for 24 hours.
Temperature coefficient	±0.1 percent per °C, over entire frequency range.
Jitter	±0.025 percent cycle-to-cycle stability.
Output voltages (simultaneous)	(a) 0 to 32.5 V p-p open circuit at 50Ω output terminals; or 0 to 30 V p-p into 600Ω load; or 0 to 10 V p-p into 50Ω load. Short circuit output current ±100 mA. (b) 5 V p-p at 50Ω output impedance. (c) Sync pulse: At least -10 V into open circuit; less than 5 us duration. (Note: All outputs may be shorted without damage to instrument.)
Accuracy (amplitude)	
As a function of frequency	Amplitude variation less than 0.1 dB 0.0015 Hz to 100 kHz and 0.5 dB 100 kHz to 1 MHz.
Peak-to-peak	±1 percent for 5 V outputs; ±1 percent for 32.5 V output at maximum gain; ±10 percent for 1 V output.
Stability (amplitude)	
At room temperature	±0.05 percent of maximum p-p values for 10 minutes; ±0.25 percent of maximum p-p values for 24 hours.

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G01 - Continued

Temperature coefficient	±0.1 percent per °C, over entire frequency range.
DC offset	±0.05 percent of maximum p-p amplitude for 10 minutes; ±0.25 percent of maximum p-p amplitude for 24 hours.
Time symmetry	99.5 percent from 0.0015 Hz to 100 kHz; 99 percent from 100 kHz to 1 MHz.
Linearity - distortion	
Symmetry	±1 percent of maximum p-p amplitude referenced to ground.
Distortion	Less than 0.5 percent from 0.0015 Hz to 10 kHz; 1.0 percent from 10 to 100 kHz; 2.0 percent from 100 kHz to 1 MHz.
Temperature	All specifications listed, except stability, are for 25° ±5°C. For operation from 0° to +55°C derate all specifications by a factor of 2.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Wavetek Model *111.

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ITEM G02

GENERATOR, SINE WAVE
(Obsolescent)

TYPE OF EQUIPMENT	Sine wave generator.
FUNCTION PERFORMED	Provide accurate (stable) variable frequency sinusoidal voltage.
SPECIFICATIONS	
Frequency	
Range	0.001 Hz to 100 kHz.
Calibration accuracy	±1 percent from 1 Hz to 10 kHz; ±3 percent from 0.01 Hz to 100 kHz; ±30 percent from 0.001 to 0.01 Hz.
Stability (after initial warm-up)	Short-term drift, less than 0.01 percent; drift per hour, less than 0.05 percent; drift for ±10 percent change in line voltage is less than 0.05 percent from 1 Hz to 10 kHz, ±0.2 percent from 0.001 Hz to 100 kHz.
Output	
Power	100 mW maximum.
Voltage	
As a function of load	10 Vrms open circuit or 1 kΩ load; 6 Vrms across 600Ω load.
As a function of frequency	Constant within ±0.25 dB from 1 Hz to 10 kHz, within ±0.75 dB from 0.01 Hz to 100 kHz, within ±3 dB from 0.001 to 0.01 Hz. All at constant line voltage. Amplitude is constant within ±0.05 dB for a ±10 percent change in line voltage at any generator output frequency.
Impedance	Nominal 1 kΩ at 5 V position of rms volts control, dropping to low values at control extremes.
DC component	Nominal zero volts.
Harmonic distortion (rms)	Less than 0.1 percent at any output level setting over frequency range from 4 Hz to 10 kHz; 3 percent at 0.02 Hz and 100 kHz; and approximately 10 percent at 0.001 Hz.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Krohn-Hite Model 440-AR.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Wavetek Model 111.
Exceptions	Harmonic distortion: ±0.5 percent from 0.0015 to 10 kHz; 1.0 percent from 10 to 100 kHz.

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ITEM G03

GENERATOR, SINE WAVE

TYPE OF EQUIPMENT	Sine wave generator.
FUNCTION PERFORMED	A variable frequency relatively constant amplitude, low distortion sine wave signal source.
SPECIFICATIONS	
Frequency	
Range	20 to 100,000 Hz.
Accuracy (stability included)	±2 percent (20 to 10,000 Hz) ±3 percent (10 to 100 kHz) (Includes all causes of drift).
Voltage	
Accuracy	±1 dB referenced to 1,000 Hz over entire frequency range.
Distortion	Less than 0.5 percent independent of load.
Power	
Output	160 mW (10 Vrms across 600Ω).
Hum and noise	More than 60 dB below rated output.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 200 CD
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AT NOTED	
Item G01	Wavetek Model 111.
Exception	Harmonic distortion 10 to 100 kHz is 0.1 to 3.0 percent.

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ITEM G04

GENERATOR, SINE WAVE

TYPE OF EQUIPMENT	Sine wave generator.
FUNCTION PERFORMED	A variable frequency constant amplitude sine wave signal source.
SPECIFICATIONS	
Frequency	
Range	10 Hz to 10 MHz.
Accuracy	±2 percent of frequency setting.
Output voltage	
Range	Adjustable from 0 to 5 Vrms into a 50Ω load.
Frequency response	Flat within 2 percent of amplitude for the entire frequency range.
Distortion	Less than 1 percent.
Noise	Less than 0.1 percent of signal voltage.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Hewlett-Packard Model 652A test oscillator
Exceptions	Frequency accuracy ±2 percent, 100 Hz to 1 MHz. ±3 percent, 10 to 100 Hz, 1 to 10 MHz. Voltage range Adjustable 0.1 mV full scale to 3.0 V full scale into 50 or 600 Ω load. Frequency response ±2 percent (100 Hz to 1 MHz). ±3 percent (10 to 100 Hz). ±4 percent (1 to 10 MHz). Noise and hum Less than 0.05 percent of maximum rated output. Special features Output is very flat over the frequency range. Panel meter to be used in setting the output voltage in Vrms or dB.

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ITEM G05

GENERATOR, FUNCTION

TYPE OF EQUIPMENT	Function generator.
FUNCTION PERFORMED	Provides accurate (stable) variable frequency sine, square, triangle, and ramp waveforms.

SPECIFICATIONS

Frequency	
Range	0.001 Hz to 1 MHz.
Voltage controlled generator (VCG)	Up to 100:1 frequency ratio for input from -0.45 V to +4.5 V. 20:1 frequency ratio for input from -0.25 V to +4.5 V (0.5 V input change for a change of one in the first significant digit, ±1 percent). Input impedance 10 kΩ.
Electronic frequency vernier	One turn for a change of at least one in the second significant digit.
VCG bandwidth	100 KHz.
Accuracy (frequency)	
Selected frequency (Sf)	±1 percent of Sf 0.001 Hz to 100 KHz; 2 percent of Sf 100 kHz to 1 MHz.
VCG linearity (0.001 Hz to 1 MHz)	±0.1 percent frequency versus input voltage (frequency error 0.1 percent of total deviation-best straight line method).
Stability (frequency)	
Short term	Drift less than ±0.05 percent of setting for 10 minutes.
Long term	Drift less than ±0.25 percent of setting for 24 hours.
Jitter	±0.025 percent cycle to cycle stability.
Output	
Selectable amplitude voltages (all waveforms)	0.03 to 30 Vp-p (15 V ramp) into open circuit; or 0.03 to 30 Vp-p (15 V ramp) into 600Ω load; or 0.03 to 10 Vp-p (5 V ramp) into 50Ω load. (Short circuit output current ±100 mA).
Fixed amplitude voltages (simultaneous)	(a) Square wave, 5 Vp-p into open circuit. (b) Sine wave, 5 Vp-p into open circuit. (c) Triangle wave, 5 Vp-p into open circuit. (d) Ramp, 0 to -2.5 V into open circuit; 50 percent duty cycle. (Note: All outputs may be shorted to signal ground without damage to the instrument).
Impedance (all terminals)	Nominal, 50Ω; 600Ω available by removing jumper wire or selectec by front panel switch.

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G05 - Continued

DC offset	±5 Vdc on selectable output. Offset plus signal limited to ±15 V peak, or ±100 mA into 50Ω load.
Accuracy (amplitude)	
Frequency response	Amplitude change with frequency less than 0.1 dB, 0.001 Hz to 100 kHz; less than 0.5 dB, 100 kHz to 1 MHz.
Peak-to-peak	±1 percent for fixed amplitude outputs; ±1 percent for 0.03 to 30 Vp-p output into 600Ω load at maximum gain; ±1 percent for 0.03 to 10 Vp-p output into 50Ω load.
Stability (amplitude)	
Short term	±0.05 percent of maximum peak-to-peak values for 10 minutes.
Long term	±0.25 percent of maximum peak-to-peak values for 24 hours.
DC offset	±0.05 percent of maximum peak-to-peak amplitude for 10 minutes; ±0.25 percent of maximum peak-to-peak amplitude for 24 hours.
Symmetry	
Amplitude	All waveforms except ramp, within ±1 percent of maximum peak-to-peak amplitude referenced to ground (with dc offset control in calibration position).
Time	All waveforms, 99.5 percent from 0.001 Hz to 100 kHz; 99 percent from 100 kHz to 1 MHz.
Linearity-distortion	
Sine wave distortion (maximum)	0.5 percent from 0.001 Hz to 10 kHz; 1.0 percent from 10 kHz to 100 kHz; 2.0 percent from 100 kHz to 600 kHz; 3.0 percent from 600 kHz to 1 MHz.
Triangle and ramp linearity (minimum)	99 percent from 0.001 Hz to 100 kHz; 95 percent from 100 kHz to 1 MHz.
Ramp fall time	250 ns or less.
Square wave rise and fall time	Fixed amplitude output, 15 ns or less; selectable amplitude output, 100 ns or less.
Total aberrations	5 percent or less (overshoot, preshoot, etc.).
Tilt	0.5 percent or less.
Temperature	All specifications listed, except stability, are for 25° ±5°C. For operation from 0° to 55°C, derate all specifications by a factor of 2.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Wavetek, Model *113. IEC, Model F-71.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Wavetek, Model 111.

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G05 - Continued

Exceptions - - - - - - - - - - - - - - -

Frequency range
0.0015 Hz to 1 MHz.

VCG linearity (0.0015 Hz to 100 kHz)
±0.1 percent frequency versus input voltage frequency error 0.1 percent of total deviation-best straight line method) from 10 to 100 percent of maximum dial frequency.

Selectable amplitude output voltages (all waveforms)
0 to 30 Vp-p into open circuit; or
0 to 30 Vp-p into 600Ω load; or
0 to 10 Vp-p into 50Ω load.

Fixed amplitude output voltages (simultaneous)
(a) Square wave, 0.5 Vp-p into 50Ω load.
(b) Square wave, 5 Vp-p into open circuit.
(c) Sine wave, 5 Vp-p into open circuit.
(d) Triangle wave, 5 Vp-p into open circuit.
(e) Sync pulse: At least -10 V into open circuit; less than 5 us duration.

DC offset control
Requires internal changes and considerations.

Peak-to-peak accuracy (amplitude)
±1 percent for 5 Vp-p fixed amplitude outputs;
±1 percent for 0 to 30 Vp-p output into 600Ω load at maximum gain;
±10 percent for 0.5 Vp-p fixed amplitude output.

Sine wave distortion (maximum)
0.5 percent from 0.005 Hz to 10 kHz;
1.0 percent from 10 to 100 kHz;
2.0 percent from 100 kHz to 1 MHz.

Triangle linearity (minimum)
99 percent from 0.005 Hz to 100 kHz;
95 percent from 100 kHz to 1 MHz.

Ramp fall time
Not applicable.

Square wave rise and fall time
0.5 Vp-p fixed amplitude output less than 10 ns into 50Ω load.
5 Vp-p fixed amplitude output less than 15 ns;
0 to 30 Vp-p selectable amplitude output less than 100 ns.

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ITEM C06

GENERATOR, FUNCTION

TYPE OF EQUIPMENT	Function generator.
FUNCTION PERFORMED	Provides accurate (stable) variable frequency sine, square, triangle, positive and negative ramps and linear ramp waveforms.

SPECIFICATIONS

Modes	
External voltage controlled frequency	Frequency ratio 1:1000 for inputs -15 mV to -15 V.
Control	Frequency determined by dial and multiplier with ±5 percent change from dial setting with an external voltage.
Pulse	Chain of pulses generated when waveform dial selector placed at square wave. Pulse widths from 200 ns to 5 s generated with predetermined frequency controlled by sweep time control.
Trigger	Applying a pulse train to trigger input produces a complete cycle of dial frequency with repetition rate of pulse train. Pressing manual trigger produces one cycle of dial frequency.
Gate	The output signal continues as long as level at trigger input is maintained or manual trigger button remains depressed.
Triggered sweep hold	Output sweeps once and holds highest frequency when trigger input is pulsed or manual trigger button is pressed.
Triggered sweep burst	When pulse is applied to trigger input or manual trigger button is pressed, one sweep will occur with final cycle stopping at start level.
Triggered tone burst	When signal is applied to trigger input or manual trigger button is pressed, output provides fixed frequency burst with length of burst determined by sweep time control.
Continuous sweep	Output sweeps between frequencies set by multiplier and sweep start and width controls at a rate controlled by sweep time. Maximum frequency-sweep ratio 1:1000.
Frequency	
Range (sine, square, triangle and ramp waveforms)	0.002 Hz to 3 MHz.
Range (linear ramp)	0.1 Hz to 100 kHz.
Accuracy (frequency)	
Selected frequency (Sf)	±5 percent full scale Sf 0.002 to 0.03 Hz; ±2 percent full scale Sf 0.03 Hz to 3 MHz.
EX VC and internal sweep	±5 percent full scale from 0.0003 to 0.0003 Hz; ±2 percent full scale from 0.0003 Hz to 3 MHz.
Stability (frequency)	
Short term	±0.05 percent in 10 minutes.
Long term	±0.25 percent in 24 hou

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G06 - Continued

Line change	±0.1 percent for ±10 percent change in line voltage.
Jitter	±0.025 percent cycle-to-cycle stability.
Inputs	
Trigger	1 V into 10 kΩ ac coupled.
Gate	1 V into 10 kΩ at trigger input.
Output	
Amplitude (all waveforms)	0 to 20 Vp-p (ramp = 10 V) into open circuit; or 0 to 10 Vp-p (ramp = 5 V) into 50Ω.
Linear ramp	0 to 10 v peak into open circuit; or 0 to 5 v peak into 200Ω.
Impedance (sine, square, ramp, pulse and triangle)	50Ω.
Impedance (linear ramp)	200Ω.
DC offset (sine, ramp, triangle, square, and pulse)	±5 V into open circuit dc offset plus signal limited to ±10 V.
DC offset (linear ramp)	0 to 10 v peak into open circuit.
Stability (amplitude)	
DC offset drift	5 mV/°C
Symmetry	
Time (square, triangle)	99 percent from 0.002 Hz to 100 kHz.
Ramp duration	47 percent of period.
Linearity-distortion	
Sine wave distortion (maximum output)	0.5 percent from 0.02 Hz to 100 kHz; 1 percent from 0.002 to 0.02 Hz; 3 percent from 100 kHz to 3 MHz.
Sine wave flatness	0.05 dB from 0.002 Hz to 100 kHz; 0.15 dB from 100 kHz to 3 MHz.
Triangle and ramp linearity	99 percent from 0.002 Hz to 100 kHz; 95 percent from 100 kHz to 1 MHz.
Ramp fall time	30 ns.
Square wave rise and fall time	30 ns.
Square wave total aberrations	5 percent with 50Ω load.
Linear ramp duration	99 percent from 0.1 Hz to 10 kHz; 90 percent from 10 to 100 kHz.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Krohn-Hite, Model 5200. IEC, Model F-34 and/or F-74.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

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ITEM G07

GENERATOR, POWER SINE WAVE

TYPE OF EQUIPMENT	Power sine wave generator.
FUNCTION PERFORMED	Variable voltage and frequency ac power source.

SPECIFICATIONS

Output	
Voltage ranges	Adjustable: 0 to 32 Vac, 0 to 130 Vac, or 0 to 260 Vac.
Wave-shape	Sine wave.
Power	0 to 200 VA.
Power factor	Full current rating into 0.7 lagging; or 0.7 leading power factor loads.
Frequency range	45 Hz to 5 kHz at rated power, 5 to 10 kHz at one half rated power.
Distortion	Less than 1.0 percent (45 Hz to 5 kHz), less than 2.0 percent (5 to 10 kHz).
Line regulation	±0.25 percent for an input voltage change of ±9 percent.
Load regulation	±1 percent no load to full load 45 Hz to 5 kHz.
Response time	50 us maximum.
Short circuit protection	Yes, with automatic recovery when short is removed.
Input power	115 ±10 Vac or 230 ±20 Vac, 47 to 63 Hz single phase.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Elgar Model 201 with variable frequency plug in, Elgar Model 401-V.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Elgar Model 251.
Exceptions	Output power rating is 250 VA.
Manufacturer	Weston Rotek Model 146 AV 10.
Exceptions	Many of the specifications are better than the above specifications (output distortion, output voltage, and frequency range and load regulation) however, only the specifications which adversely affects the performance are listed below: Output power: 10 W maximum (10 V to maximum and 380 to 2000 Hz), less for other ranges (see manufacturers specification).

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ITEM 608

GENERATOR, SINE WAVE

TYPE OF EQUIPMENT	Sine wave generator.
FUNCTION PERFORMED	Provide accurate (stable) variable frequency sinusoidal voltage.

SPECIFICATIONS

Range	0.1 Hz to 100 kHz.
Calibration accuracy	±0.1 percent from 10 Hz to 9.99 kHz; ±0.25 percent at 50 kHz; ±0.5 percent at 100 kHz; ±1 percent at 1 Hz.
Stability vs time	±0.01 percent from 0.1 Hz to 100 kHz within one hour; ±0.01 percent from 10 Hz to 9.99 kHz and ±0.1 percent at 1 Hz and 100 kHz within 10 hours.
Stability vs line	±0.005 percent for 10% line voltage change.
Stability vs load	±0.005 percent from 1 Hz to 10 kHz and ±0.05 percent from 0.1 Hz to 100 kHz, no load to full load.
Stability vs temperature	±0.005 percent per degree C from 10 Hz to 9.99 kHz, and 0.05 percent per degree C at 0.1 Hz and 100 kHz.
Control	Direct settings of three significant digits with a frequency vernier of infinite resolution to adjust between direct settings.
Output (Main)	
Connections	200 and 600 Ω outputs.
Power	1[illegible] mW.
Voltage (maximum)	10 Vrms open circuits; 5 Vrms across 200 and 600 Ω.
Frequency response (no load or resistive load)	±0.01 dB from 20 Hz to 20 kHz; ±0.05 dB from 1 Hz to 100 kHz; ±0.5 dB to 0.1 Hz.
Harmonic distortion	< 0.01 percent from 10 Hz to 20 kHz rising to 0.05 percent at 50 kHz and 0.1 percent at 0.1 Hz and 100 kHz.
Hum and noise	< 0.01 percent.
Amplitude stability vs time	±0.02 percent from 20 Hz to 20 kHz and ±0.1 percent from 1Hz to 100 kHz within any period of 10 hours or less.

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ITEM G08 - Continued

Amplitude stability vs line	±0.1 percent from 1 Hz to 100 kHz for 10 percent line voltage changes.
Amplitude stability vs temperature	±0.1 percent per degree C from 1 Hz to 100 kHz.
Control	The output voltage vernier has infinite resolution.
Output (Quadrature)	
Connections	200 and 600 Ω outputs.
Voltage	Fixed at 10 Vrms ± 2 percent.
Amplitude stability	Approximately twice that at main output for time, line, and temperature.
Harmonic distortion	Approximately twice the main output values.
Frequency response	±0.1 dB from 20 Hz to 20 kHz and ±0.5 dB from 0.1 Hz to 100 kHz.
Floating operation	A switch disconnects the signal ground from chassis ground allowing balanced output operation. Performance is slightly reduced above 10 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Kron-Hite Model 4001.

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ITEM G09

GENERATOR, POWER SINE WAVE

TYPE OF EQUIPMENT	Power sine wave generator.
FUNCTION PERFORMED	Provide precise variable frequency, single or 3 phase signal source.
SPECIFICATIONS	
Output power	0 to 1,500 VA; 0 to 500 VA/ phase.
Power factor	Unity to 0.7.
Output voltage	Adjustable 0 to 130 Vac line to neutral and 0 to 130 Vac line to line.
Frequency range	45 Hz to 6 kHz.
Frequency accuracy	±1 percent of set value.
Distortion	< 1 percent 45 Hz to 6 kHz and < 0.5 percent 400 to 800 Hz.
Load regulation	Line to neutral < 1 percent; line to line balanced load, < 1 percent 45 to 800 Hz, < 2 percent 800 Hz to 6 kHz; line to line unbalanced load, < 2 percent 45 to 800 Hz, < 3 percent 800 Hz to 6 kHz.
Line regulation	0.5 percent for ±10 percent line change.
Hum and noise	60 dB below rated output.
Output short circuit protection	Output is completely short circuit protected.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Elgar 1503 power source with 400 V or 403 V plug-in oscillator.

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ITEM H01

GENERATOR, SQUARE WAVE

TYPE OF EQUIPMENT	Square wave generator.
FUNCTION PERFORMED	Provides accurate variable frequency square wave and sync pulses.

SPECIFICATIONS

Frequency	
Range	0.0015 Hz to 1 MHz.
Voltage - controlled generator (VCG)	4.75 V input for 20:1 frequency ratio. Input impedance 10 kΩ.
Electronic freqeuncy vernier	1 turn for 1 minor dial division.
VCG bandwidth	100 kHz.
Accuracy (frequency)	
Dial	±1 percent full scale from 0.0015 Hz to 100 kHz; ±2 percent full scale from 100 kHz to 1 MHz.
VCG linearity (0.0015 to 100 Hz)	±0.1 percent frequency versus input voltage (frequency error 0.1 percent of total deviation - best straight line method) from 10 to 100 percent of maximum dial frequency.
Stability (frequency)	
Room temperature	Drift less than ±0.05 percent of setting for 10 minutes. Drift less than ±0.25 percent of setting for 24 hours.
Temperature coefficient	±0.1 percent per °C, over entire frequency range.
Jitter	±0.025 percent cycle-to-cycle stability.
Output voltages (simultaneous)	(a) 0 to 32.5 V p-p open circuit at 50Ω output terminals; 0 to 30 V p-p into 600Ω load; or 0 to 10 V p-p into 50Ω load. (Short circuit output current ±100 mA.) (b) 1 V p-p, 5 ns rise time at 50Ω output impedance. (c) 5 V p-p, 15 ns rise time at 50Ω output impedance. (d) Sync pulse: At least - 10 V into open circuit; less than 5 us ouration. (Note: All outputs may be shorted without damage to instrument.)
Accuracy (amplitude)	
Variation as a function of frequency	Less than 0.1 dB 0.0015 Hz to 100 kHz; and less than 0.5 dB 100 kHz to 1 MHz.
Peak-to-peak	±1 percent for 5 V outputs; ±1 percent for 32.5 V output at maximum gain; ±10 percent for 1 V output.
Stability (amplitude)	

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H01 - Continued

Room temperature	±0.05 percent of maximum p-p values for 10 minutes; ±0.25 percent of maximum p-p values for 24 hours.
Temperature coefficient	±0.1 percent per °C, over entire frequency range.
DC offset (as a function of time)	±0.05 percent of maximum p-p amplitude for a period of 10 minutes; ±0.25 percent of maximum p-p amplitude for a period of 24 hours.
Time symmetry	99.5 percent from 0.0015 Hz to 100 kHz; 99 percent from 100 kHz to 1 MHz.
Linearity - distortion	
Symmetry	Within ±1 percent of maximum p-p amplitude referenced to ground.
Rise and fall time	1 V output less than 5 ns; 5 V output less than 15 ns; 32.5 V output less than 100 ns.
Total aberrations	Less than 5 percent (overshoot, preshoot, etc.).
Tilt	Less than 0.5 percent.
Temperature	All specifications listed, except stability are for 25° ±5°C. For operation from 0° to 55°C, derate all specifications by a factor of 2.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Wavetek Model *111.

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ITEM H02

GENERATOR, SQUARE WAVE
(Obsolescent)

TYPE OF EQUIPMENT	Square wave generator.
FUNCTION PERFORMED	Provide accurate (stable) variable frequency square wave signals.
SPECIFICATIONS	
Frequency	
Range	0.001 Hz to 100 kHz.
Calibration accuracy	±1 percent from 1 Hz to 10 kHz, ±3 percent from 0.01 Hz to 100 kHz, ±30 percent from 0.001 Hz to 0.01 kHz.
Stability (after initial warm-up)	Short-term drift, less than 0.01 percent; drift per hour, less than 0.05 percent; drift for ±10 percent change in line voltage, less than ±0.05 percent from 1 Hz to 10 kHz; ±0.2 percent from 0.001 Hz to 100 kHz.
Output	
Voltage	10 V p-p open circuit; 5 V p-p across 1,500Ω load.
Amplitude	Constant within ±1 percent from 0.001 Hz to 100 kHz at constant line voltage. Within ±1 percent for a 10 percent change in line voltage at any frequency.
Internal impedance	Nominal, 1,500Ω.
DC component	Nominal zero volts.
Shape	Rise time less than 0.5 us. Symmetrical within 8 percent; flat top with no droop, or overshoot at any frequency; positive and negative half-cycle amplitudes equal within ±10 percent.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Kronn-Hite Model 440-AR.
Item	H01.

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ITEM J01

GENERATOR, PULSE

TYPE OF EQUIPMENT	Pulse generator.
FUNCTION PERFORMED	Generation of pulse trains with controlled parameters.
SPECIFICATIONS	
Pulse repetition frequency	
Single pulse	5 Hz to 50 MHz.
Double pulse	5 Hz to 25 MHz.
Pulse polarity	Complementary positive and negative pulses (i.e., Q and $\bar{Q}$).
Pulse amplitude	Each output independently variable from 10 mV to 10 V into a 50Ω load.
Pulse width	Adjustable from 10 ns to 5 ms.
Rise and fall time	5 ns to 500 µs with a common rise/fall control having at least five decade ranges and separate verniers for independent control of rise and fall time in each range. The ratio of rise to fall time shall be at least 15 to 1.
Aberrations	Overshoot, undershoot, droop and ringing shall not exceed 3 percent at amplitudes above 0.3 V.
Base line	Adjustable from -10 Vdc to +2 Vdc for the positive pulse output and +10 Vdc to -2 Vdc for the negative pulse output.
Jitter	Repetition rate, delay and width jitter not greater than 0.05 percent.
Pulse trigger modes	Internal, external and single cycle manual.
Gated operation	Synchronous and asynchronous gating. Capable of being gated from a signal that can supply at least 3 volts into 1 kΩ impedance.
Sync output	At least 2 volt amplitude of either polarity and capable of working into 50Ω impedance.
Pulse delay	Variable over 75 percent of pulse period up to 1 MHz repetition rate, 40 percent 1 to 20 MHz and 10 percent at 50 MHz.
Output protection	The output stages shall not be damaged by any combination of front panel control settings, open or short circuits, or reverse voltages up to 10 volts.
EQUIPMENT MEETING ALL SPECIFICATIONS	

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J01 - Continued

Manufacturer	Datapulse Model 110B. IEC, Model P-25.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Manufacturer	Datapulse Model 110 and Model 110A.
Exceptions	(a) Maximum frequency is 40 MHz. (b) Aberration does not exceed 3 percent with outputs above 0.5 volt. (c) Delay and width maximum jitter is 0.1 percent.
Manufacturer	Datapulse Model 111.
Exceptions	(a) Maximum frequency is 40 MHz. (b) Aberration does not exceed 3 percent with outputs above 0.5 volt. (c) Delay and width maximum jitter is 0.1 percent. (d) Maximum rise and fall time is 500 ns. (e) Maximum output is 5 volts.

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ITEM J02

GENERATOR, PULSE

TYPE OF EQUIPMENT	Pulse generator.
FUNCTION PERFORMED	Generation of rectangular pulse of known and controllable constants.
SPECIFICATIONS	
Pulse	
Repetition rate (PRR)	0.3 Hz to 10 MHz variable with at least 5 ranges.
Duty cycle	90 percent from 0.3 Hz to 1 MHz. 50 percent from 1 MHz to 10 MHz.
Pulse shape	(Limits assume proper termination).
Rise time (t_R) and fall time (t_F)	10 ns to 2 s with at least 5 ranges and verniers allow independent control of leading and trailing edges on each range.
Width	30 ns to 3 s with at least 4 ranges.
Aberrations	Preshoot, overshoot, ringing: Maximum 5 percent of pulse amplitude.
Amplitude	Variable 0.02 to 5 V into 50Ω, with at least 7 steps of attenuation. Positive or negative pulse polarity.
External trigger	
Input signal requirements	A 2 Vp-p sine wave or square wave of either polarity and amplitude of at least 1 V up to 10 MHz.
Input impedance	1 kΩ.
Manual trigger	
Function	Pushbutton for single pulse.
Trigger output	
Amplitude	+2 V or greater across 50Ω.
Width	15 ±5 ns.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 8002A. IEC, Model P-24.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

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ITEM J03

GENERATOR, WORD

TYPE OF EQUIPMENT	Word generator.
FUNCTION PERFORMED	A controlled synchronous digital logic generator for complex test programs.
SPECIFICATIONS	
Frequency stepping	(a) 1 kHz to 10 MHz in 1-3-10. (b) Fine control for continuous coverage.
Program bit-times	(a) 16. (b) Selectable to limit for quads or octets.
Program outputs	10.
Output voltage	+2.5 V minimum into 50Ω.
Output current	At least 50 mA.
Program length	Selectable.
Ranges	0-3, 4-7, 8-11, 12-15, 0-7, 8-15, or 0-15.
Delay and width	(a) 30 ns to 100 us continuous. (b) Delay plus width time up to 50 percent of clock interval.
Jitter	0.02 percent ±30 ps bit-to-bit, or less.
Repeat channels	(a) Two channels for step repeats. (b) Two channels for even/odd pairs. (c) One for quad. (d) One for octet.
Range	1 us to 1 s in 1-10 steps with continuous coverage control.
Clock modes	
Single step	Allows manual advancing from step to step.
Gate off	Clock free runs at selected frequency with set and reset controls.
Gate on	External trigger lets clock run for duration of gate signal.
Single pass	External or manual switch initiates clock for 1 word.
Multiple pass	External gate controls multiple of complete words.
Sync output	Pre-trigger at clock frequency.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	E. H. Research Labs, Inc., Model 1623 word generator.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTION AS NOTED	

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ITEM J04

GENERATOR, PULSE

TYPE OF EQUIPMENT	Pulse generator.
FUNCTION PERFORMED	Generator of single pulses on pulse trains with controlled parameters.
SPECIFICATIONS	
Modes of operation	Internal, external trigger, gate, manual (one-shot pushbutton) single pulse (pulse 1 only), delayed pulse (pulse 2 only), double pulse.
Frequency data	
Pulse repetition rate (PRR)	1 Hz to 25 MHz (at reduced amplitudes). Higher effective (PRR) in double pulse mode. At least 3 MHz repetition rate at 100 V. Continuously variable.
Frequency jitter	0.1 percent of period +0.5 ns.
Pulse width data	
Range	20 ns to 500 ms less than 25 V; 20 ns to 5 ms @ 50 V; 30 ns to 0.1 ms @ 100 V.
Variation	Continuously variable; independent range and vernier control for pulse 1 and pulse 2.
Stability	0.1 percent of setting +0.5 ns.
Pulse delay data	
Range	15 ns to 500 ms (pulse 2 position with respect to leading edge of pulse 1). 45 ns to 500 ms, pulse 2 position with respect to sync.
Variation	Continuously variable.
Stability	0.1 percent of setting +0.5 ns.
Control	(Double pulse mode) controls pulse 1 to pulse 2 spacing.
Delayed pulse mode	Controls delay of output with respect to sync out and trigger in pulse 2 only output.
Input data	
Gate/trigger input	May be gated or triggered at rates to 25 MHz.
Internal sync	Internal clock may be synchronized by an external pulse; frequency ratio to 100:1 may be used to sync.
Threshold	-2 V to +2 V; continuously variable through zero.
Sensitivity	Less than 0.1 V, 5 ns or wider input.
Impedance	Low (50Ω) or high (225Ω) switch selector.
Slope	Positive or negative; switch selector.
Delays	Trigger to sync, less than 15 ns; trigger to output less than 60 ns; sync to output, 45 ns.

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ITEM J04

GENERATOR, PULSE

TYPE OF EQUIPMENT	Pulse generator.
FUNCTION PERFORMED	Generator of single pulses on pulse trains with controlled parameters.
SPECIFICATIONS	
Modes of operation	Internal, external trigger, gate, manual (one-shot pushbutton) single pulse (pulse 1 only), delayed pulse (pulse 2 only), double pulse.
Frequency data	
Pulse repetition rate (PRR)	1 Hz to 25 MHz (at reduced amplitudes). Higher effective (PRR) in double pulse mode. At least 3 MHz repetition rate at 100 V. Continuously variable.
Frequency jitter	0.1 percent of period +0.5 ns.
Pulse width data	
Range	20 ns to 500 ms less than 25 V; 20 ns to 5 ms @ 50 V; 30 ns to 0.1 ms @ 100 V.
Variation	Continuously variable; independent range and vernier control for pulse 1 and pulse 2.
Stability	0.1 percent of setting +0.5 ns.
Pulse delay data	
Range	15 ns to 500 ms (pulse 2 position with respect to leading edge of pulse 1). 45 ns to 500 ms, pulse 2 position with respect to sync.
Variation	Continuously variable.
Stability	0.1 percent of setting +0.5 ns.
Control	(Double pulse mode) controls pulse 1 to pulse 2 spacing.
Delayed pulse mode	Controls delay of output with respect to sync out and trigger in pulse 2 only output.
Input data	
Gate/trigger input	May be gated or triggered at rates to 25 MHz.
Internal sync	Internal clock may be synchronized by an external pulse; frequency ratio to 100:1 may be used to sync.
Threshold	-2 V to +2 V; continuously variable through zero.
Sensitivity	Less than 0.1 V, 5 ns or wider input.
Impedance	Low (50Ω) or high (225Ω) switch selector.
Slope	Positive or negative: switch selector.
Delays	Trigger to sync, less than 15 ns; trigger to output less than 60 ns; sync to output, 45 ns.

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ITEM J05

GENERATOR, PULSE

TYPE OF EQUIPMENT	Pulse Generator.
FUNCTION PERFORMED	A versatile pulse generator with very fast variable transition times and repetition rate up to 100 MHz.
SPECIFICATIONS	
Mode of operation	External triggering and synchronous gating. Trigger level adjustable for all externally controlled modes with the slope polarity selectable.
Pulse characteristics (50Ω source and load impedance).	
Transition times	<2 ns to 250 us, three ranges, common for both transition times. Independent vernier adjustments for leading and tailing edges up to maximum ratios of 1:50 or 50:1.
Linearity	Maximum amplitude deviation from a straight line between 10 and 90 percent points ≤ 5 percent of pulse amplitude.
Preshoot, overshoot, ringing	< ± 5 percent of pulse amplitude.
Pulse width	< 5 ns to 50 ms in 5 ranges.
Width jitter	Less than 0.1 percent.
Maximum duty cycle	Normal >50 percent; complementary approximately 100 percent.
Amplitude	5 V maximum (10 V across open circuit).
Pulse output	Polarity selectable; normal, complement or symetrical to ground.
Source impedance	50 ± 4Ω shunted by typically 10 pF.
DC offset	± 4 V across 50Ω load.
Pulse delay	Less than 30 ns to 50 ms with respect to trigger output.
Pulse delay jitter	<0.1 percent on any delay setting.
Repetition rate and trigger	10 Hz to 100 MHz in 5 ranges (continuous adjustment within ranges).
Period jitter	Less than 0.1 percent.
Double pulse	Up to 50 MHz (representing an output pulse rate of 100 MHz).
Trigger output	Greater than + 1 V across 50Ω, 4 ± 2 ns wide.
External triggering	0 to 100 MHz.
Delay	15 ns between trigger input and output.
Manual	Single pulse mode.
Synchronous gating	Gating signal turns generator "on". Last pulse is completed even if gate ends during pulse.

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KO1

GENERATOR, TIME-MARK

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Time-mark generator.

FUNCTION PERFORMED - - - - - - - - - - - - - A controlled frequency, constant amplitude, sine wave and marked interval source.

SPECIFICATIONS

Frequency
Range
Marker output - - - - - - - - - - - - - 0.1 us to 5 s in 1-5-10 sequence.
Sine wave output - - - - - - - - - - - - 2, 5, 10, 20, and 50 ns.
Trigger output - - - - - - - - - - - - - 1 us to 1 s in 1-10 sequence.
Accuracy - - - - - - - - - - - - - - - - Crystal controlled oscillator of 10 MHz ±0.001 percent from 20° to 30°C, 10 MHz ±0.002 percent from 0° to 50°C within 5 minutes warmup.

Output voltage
Range
Marker output - - - - - - - - - - - - - 0.5 V minimum peak amplitude into 50Ω.
Sine wave output - - - - - - - - - - - - 0.3 V minimum peak-to-peak amplitude into 50Ω.
Trigger output - - - - - - - - - - - - - 0.4 V minimum amplitude into 50Ω.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Tektronix - Type 184
Tektronix - Type 2901

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard Model 266A.
Exceptions - - - - - - - - - - - - - - - - Accuracy -
±0.005 percent, 0° to +55°C
±0.002 percent at 25°C after 30-minute warmup.
Sine wave output - 2 ns and 5 ns only.

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ITEM K02

PHOTOMETER, DIGITAL

TYPE OF EQUIPMENT	Digital photometer and probe.
FUNCTION PERFORMED	
Using LED test probe	Measures luminous intensity of light sources having spectral output in the red region.
Using illuminance probe	Measures luminous intensity of light sources having spectral outputs in the yellow and green regions.
SPECIFICATIONS	
Photometer	
Range	X.1, X1, X10, X100, X1000 fc.
Stability	Within 2 percent per year.
Linearity	Within 2 percent over entire range.
System accuracy (photometer and any probe)	±5 percent of full scale, ±1 digit on any range.
Operating temperature range	-15° to +40°C.
Warm-up time	1 minute.
LED test probe	
Range	
Without test fixture adapter	0.199 to 1990 fc.
With test fixture adapter	0.0199 to 199 cd.
Maximum resolution	
Without test fixture adapter	0.001 fc.
With test fixture adapter	0.1 mcd.
Spectral response	Within 2 percent of CIE photopic curve from 600 to 710 nm.
Acceptance angle	50 percent sensitivity at ±8° off-axis.
Illuminance probe	
Range	
Without test fixture adapter	0.199 to 1990 fc.
With test fixture adapter	0.0199 to 199 cd.
Maximum resolution	
Without test fixture adapter	0.001 fc.
With test fixture adapter	0.1 mcd.
Spectral response	Within 2 percent of CIE photopic curve.
Acceptance angle	50 percent sensitivity at ±8° off-axis.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Tektronix, Incorporated Type J16 photometer used with the following probes and accessories.

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ITEM K03

GENERATOR, RANDOM NOISE

TYPE OF EQUIPMENT	Random noise generator.
FUNCTION PERFORMED	To generate random noise of Gaussian distribution of amplitudes.
SPECIFICATIONS	
General information	
Frequency	From 5 Hz to 5 MHz in 3 ranges.
Ranges	(a) 5 Hz to 20 kHz. (b) 5 Hz to 500 kHz. (c) 5 Hz to 5 MHz.
Output	
Voltage	(a) Maximum open circuit voltage is 3 V for 20 kHz range; 2 V for 500 kHz range; and 1 V for 5 MHz range. (b) Continuous control.
Impedance	(a) Source impedance is 900Ω. (b) Output is taken from 2,500Ω potentiometer. (c) Attenuated output is 200Ω.
Range	2 position switch LOW and HIGH.
Meter	Rectifier-type rms voltmeter.
Attenuator	5 position switch in 20 dB x10 steps, X1, X0.1, X0.01, X0.001, and X0.0001.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	General Radio, Model 1390-B.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM L01

BRIDGE, INDUCTANCE

TYPE OF EQUIPMENT	Inductance bridge.
FUNCTION PERFORMED	Measures inductance, dissipation factor, and quality factor of two terminal networks from 20 Hz to 20 kHz.
SPECIFICATIONS	
Inductance	
Range	0.1 uH to 1,100 H.
Accuracy	1 percent of reading.
Dissipation factor	
Range	0.0001 to 10,000.
Accuracy	10 percent of reading.
Quality factor	
Range	0.0001 to 10,000.
Accuracy	10 percent of reading.
Frequency range	20 Hz to 20 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Electro Scientific Industries Model 290A. Impedance bridge used in conjunction with 861A ac generator-detector and 840 (or 803) dc generator-detector (see C01).
Exceptions	Inductance accuracy - Worst accuracy is 0.2 percent of reading +1 dial division + 1.2 percent x F_{kHz}/Q. Below L = 1 uH accuracy is worse than 1 percent of reading. Special features - Measures both parallel and series inductance. Also this instrument satisfies the requirements of C01.
Manufacturer	General Radio Co. Model 1608A used in conjunction with a 1210C oscillator and a 1232A tuned amplifier and null-detector.
Exceptions	Dissipation factor range - 0.0005 to 2 at 1 kHz. Quality factor range - 0.5 to 2,000 at 1 kHz. Special features - Measures capacitance, resistance, inductance, and conductance.

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ITEM LO2

BRIDGE, INDUCTANCE

TYPE OF EQUIPMENT	Inductance bridge.
FUNCTION PERFORMED	Measures small values of inductance.
SPECIFICATIONS	
Inductance	
Range	0 to 300 mH.
Accuracy	±3 percent of reading.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Tektronix type 130 L-C meter.
Exceptions	Accuracy - ±3 percent of full scale.
Item	Item LO1 - Inductance bridge (equipment satisfying LO1 requirements, satisfies this specification).

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ITEM L03

BRIDGE, INDUCTANCE, RF

TYPE OF EQUIPMENT	Inductance bridge.
FUNCTION PERFORMED	Measures inductance of 2-terminal networks at radio frequencies.
SPECIFICATIONS	
Inductance	
Range	0.001 uH to 100 mH.
Accuracy	2 percent of reading.
Frequency range	0.5 to 250 MHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturers	Boonton 250A RX meter (see C02) and HP Model 250B RX meter.
Exceptions	Accuracy $\pm 0.5\ (1 + F^2\ C \times 10^{-5})$: ± 0.15 pF with F in MHz and C in pf.

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ITEM M01

MULTIMETER (VOM)

TYPE OF EQUIPMENT	Multimeter (VOM).
FUNCTION PERFORMED	Measures ac volt and dc volt/ohm/milliamperes. Used chiefly as a "troubleshooting" instrument.
SPECIFICATIONS	
DC volts	2.5 V full scale to 5,000 V full scale (minimum 6 ranges) sensitivity minimum of 20 kΩ/V full scale.
AC volts	2.5 V full scale to 5,000 V full scale (minimum 6 ranges) sensitivity minimum of 1 kΩ/V full scale.
DC resistance	Minimum of 3 ranges, 2 kΩ/200 kΩ/20 MΩ.
DC current	50 or 100 uA full scale to 10 A full scale (minimum of 5 ranges).
Accuracy	5 percent full scale ac volts 3 percent full scale dc volts 5 percent full scale dc amps 10 percent center scale Ω.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Simpson Model #260.
Manufacturer	Triplett Model 630 series.
Manufacturer	Weston Model 980 Mk II.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM M02

DIGITAL, MULTIFUNCTION METER

TYPE OF EQUIPMENT	Digital multi-function meter (integrating type).
FUNCTION PERFORMED	Measures ac, dc and ohms with ratio capability.
SPECIFICATIONS	
DC volts	(a) ±100.00 mV full scale to ±1000.00 V full scale (minimum of 5 ranges) with 20 percent overranging on all scales. (b) 4 ratio ranges X1, X10, X100, and X1000 accommodates reference voltages from ±100 mV to ±120 V.
AC volts	1.00000 V full scale to 1000.00 V full scale (minimum of 4 ranges) true rms.
Frequency	True rms response from 45 Hz to 1 MHz.
DC resistance	100.0000 Ω full scale to 10000.0 kΩ full scale (minimum of 6 ranges).
Accuracy (at +25° ±5°C) 30 days DC	(a) 1 V full scale to 1000 V full scale ±(0.008 percent of reading +0.002 percent of range). (b) 100 mV full scale ±(0.008 percent of reading +0.01 percent of range). (c) 90 days add 0.002 percent of range to (a) and (b).
DC ratio (25° ±5°C) 30 days	±(0.01 percent of reading +0.002 percent of ratio range + $\frac{\text{Y range}}{\text{Y voltage}}$ x +0.002 percent of range). * 0.003 percent for 90 days.
AC Vrms, 90 days (25° ±5°C)	(a) 45 Hz to 100 Hz. 1 mV to 500 V (0.12 percent of reading +0.03 percent of range). 500 V to 1,060 ** V (0.25 percent of reading). (b) 100 Hz to 200 Hz. 1 mV to 500 V (0.08 percent of reading +0.02 percent of range). 500 V to 1,060 ** V (0.15 percent of reading) (c) 200 Hz to 5 kHz. 1 mV to 500 V (0.04 percent of reading +0.01 percent of range). 500 V to 1060 ** V (0.1 percent of reading). (d) 5 kHz to 50 kHz. 1 mV to lesser of 500 V or V x F ≤ 10^7)*. (0.08 percent of reading +0.02 percent of range). (e) 50 kHz to 100 kHz. 10 mV to (V x F ≤ 10^7)*. (0.12 percent of reading +0.03 percent of range).

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M02- Continued

(f) 100 kHz to 200 kHz.
$50\ mV$ to $(V \times F \leq 10^7)$*.
(0.3 percent of reading +0.03 percent of range).

(g) 200 kHz to 500 kHz.
$50\ mV$ to $(V \times F \leq 10^7)$*.
(0.8 percent of reading +0.05 percent of range).

(h) 500 kHz to 1 MHz.
$50\ mV$ to $(V \times F \leq 10^7)$
(2.0 percent of reading +0.1 percent of range).

** 1060 Vrms (1,500 v peak) maximum

* $V \times F \leq 10^7$ = voltage x frequency 10^7 V x Hz.

AC ratio, 90 days (25° ±5°C) - - - - - ± (0.02 percent of reading +0.01 percent of ratio range + sum of accuracies of X and Y inputs determined from specification accuracy).

Ohms (25° ±5°C) 30 days - - - - - - - - (a) 100Ω range ±(0.01 percent of reading +0.0) percent of range).
1 kΩ to 100 kΩ range ±(0.01 percent of reading +0.002 percent of range).
1,000 kΩ range ±(0.02 percent of reading +0.002 percent of range).
10,000 kΩ range ±(0.1 percent of reading +0.002 percent of range).

(b) 90 days add 0.002 percent of range to each range above.

Ohms ratio, 30 days (25° ±5°C) - - - - ± (percent of ratio reading error + percent of ratio range error), where percent of ratio reading error is the greater percentage given below for the X and Y resistances.

0 to 100Ω	5 percent
100Ω to 1 kΩ	0.55 percent
1 to 2 kΩ	0.1 percent
2 to 9 kΩ	0.05 percent
9 to 500 kΩ	0.02 percent, or 0.01 percent for ratios between 0.95 and 1.05 if X and Y are between 10 and 500 kΩ.
500 kΩ to 5 MΩ	0.05 percent
5 to 12 MΩ	0.2 percent

% of ratio range error =

$$\left(0.04\% + \frac{\text{Y range}}{\text{Y resistance}}\right) \times 0.002\%^{**}$$

** 0.003 % for 90 days.

Temperature coefficient (0° to 50°C) - - (a) DC ± (0.0004 percent of reading +0.0003 percent of range) per °C.

(b) Ohms ±(0.0006 percent of reading +0.0003 percent of range) per °C.

(c) DC ratio ±(0.0006 percent of reading +0.0003 percent of range) per °C.

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MO2 - Continued

	(d)	AC ratio (200 Hz to 5 kHz) ±(0.002 percent of reading +0.0001 percent of ratio range) per °C.
	(e)	Ohms ratio ±(0.0009 percent of reading +0.0003 percent of ratio range) per °C.
Input impedance - - - - - - - - - - - - - -	(a)	$>10^{10}$ Ω for 100 mV, 1 and 10 Vdc ranges; 10 MΩ ±0.1 percent for 100 and 1,000 Vdc ranges.
	(b)	2 MΩ shunted by 90 ±10 pF in series with 0.1 uF (ac).
Maximum peak input voltage - - - - - - - -	(a)	±1,500 v peak between any two of the 4 input terminals of X and Y, except b and c below.
	(b)	±200 v peak between two Y input terminals.
	(c)	±200 v peak between the two low terminals and between either to guard.
	(d)	±500 v peak between chassis and guard terminal.
Sample periods - - - - - - - - - - - - - -		1/10 to 1/60 s variable.
Spurious signal rejection		
Common mode - - - - - - - - - - - - - -	(a)	Minimum of 145 dB to 1/10 s integration period.
	(b)	Minimum of 130 dB to 1/60 s integration period.
Current thru R_X - - - - - - - - - - - - - -	(a)	1 mA; 100Ω - 10 kΩ range.
	(b)	10 uA; 100 kΩ, 1,000 kΩ range.
	(c)	1 uA; 10,000 kΩ range.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - -	Hewlett-Packard Model 3450A with options 001 and 002.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

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ITEM NO1

FILTER, TUNABLE

TYPE OF EQUIPMENT	Tunable active filter, with low pass, high pass, bandpass, and band reject capability.
FUNCTION PERFORMED	Low-pass filter, high-pass filter, bandpass filter, and band reject filter.
SPECIFICATIONS	
Gain	0 ± 1 dB or -20 ± 0.2 dB reference 1 volt.
Frequency	
Selectable cutoff (-3 dB)	4 Hz to 60 kHz (±2 percent dial calibration accuracy).
High pass range	300 kHz.
Low pass range	DC.
Bandpass ratio limits	$\frac{\text{Upper frequency}}{\text{Lower frequency}}$ must be greater than 1.3.
Band reject ratio limits	$\frac{\text{Upper frequency}}{\text{Lower frequency}}$ must be greater than 2.0.
Filter type	Fourth order (four pole) Chebyshev approximation to ideal response.
Passband ripple	±0.2 dB maximum.
Attenuation	-3 dB at cutoff, -30 dB per octave initial, at least -30 dB at 1/2 Fc or 2 Fc (center frequency = Fc).
Input impedance	100 kΩ.
Output impedance	600Ω, output linearity not affected by load.
Input voltage	3 Vrms (30 Vrms with -20 dB attenuator selected).
DC component of ac voltage	100 V if ac coupling selected.
Maximum dc input if dc coupling used	±4.2 V.
Noise	Less than 100 uVrms with bandwidth of 50 kHz.
Distortion	0.25 percent in passband up to 50 kHz.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	General Radio Company Type 1952.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM P01

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A continuously variable dc voltage power source with excellent stability, regulation, load handling characteristics and low ripple.
SPECIFICATIONS	
DC output	0 to 60 V at 0 to 2.5 A.
Remotely programmable	100G/V with resolution less than 1 mV.
Regulation, standard	Line: 0.01 percent or 2 mV for 105 to 125 V line change. Load: 0.01 percent or 2 mV for no load to full load.
Ripple	Less than 500 μVrms, positive or negative ground.
Remote sensing	Provision is made for remote sensing to eliminate the effects of power output lead resistance on dc regulation.
Stability	0.05 percent +3 mV for 8 hours after warmup, under fixed line, load, and temperature conditions.
Ambient temperature range	Operating: 0° to +45°C. Nonoperating: -40° to +75°C.
Temperature coefficient	0.05 percent per °C.
Output polarity	Positive, negative, or floating, with respect to ground.
Circuit protection	Current limiting protects load and power supply. Fuses protect power supply ac input and regulated dc output lines.
Transient response	Between 200 and 300 ms.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Electronic Measurements Company Model *PO 60-2.5.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Trygon Model HR 60-2.5B.
Exceptions	Stability: ±0.05 percent or 18 mV/8 hours.

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ITEM P02

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A continuously variable dc voltage source with excellent stability, regulation, load handling capabilities and low ripple.
SPECIFICATIONS	
DC output	0 to 60 V, with resolution <1 mV, 0 to 0.13A (up to 40°C ambient).
Remotely programmable	200 Ω/V with resolution less than 1 mV.
Regulation	
Line	0.01 percent plus 2 mV for line variation of 105 to 132 Vac.
Load	0.02 percent plus 2 mV for load variations from 0 to full load.
Ripple and noise	0.5 mVrms, 1.5 mV p-p with 60 Hz input.
Remote sensing	Provision is made for remote sensing to eliminate the effect of power output lead resistance on dc regulation.
Stability	0.05%/8 hours.
Ambient temperature range	
Operating	-20° to +71°C.
Nonoperating	-55° to +85°C.
Temperature coefficient	0.05%/°C.
Output polarity	Positive, negative, or floating with respect to ground.
Circuit protection	Current limiting protects load and power supply.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Lambda Electronics Co. Model: (a) LM207Y or (b) LM208Y or (c) LM266Y.
Item	P01*.

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ITEM P03

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A 0 to 500 Vdc, 0 to 500 mA adjustable constant voltage dc power supply.

SPECIFICATIONS

DC output	
Voltage	0 to 500 Vdc
Current	0 to 500 mA
Line regulation	0.15 percent or 0.1 V for a line change from 105 to 125 Vac.
Load regulation	0.5 percent or 0.3 V for a change from no load to full load.
Ripple and noise	10 mV maximum.
Operating temperature	0° to 50°C.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturers	Lambda Model 50R Krohn-Hite Corp. Model UHR-240 Kepco Model HB525M Kepco Model 605 Electronic Measurements Model 209B Electronic Measurements Model 235AM.

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ITEM P04

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A 0 to 60 Vdc, 0 to 15 A adjustable constant voltage dc power supply.

SPECIFICATIONS

DC output	
Voltage	0 to 60 Vdc
Current	0 to 15 A.
The power supply shall have remote programming features.	
Line regulation	No greater than 60 mVdc of output change for a line change from 105 to 125 Vac.
Constant voltage load regulation	No greater than 120 mVdc for a change from no load to full load.
Ripple and noise	60 mV maximum.
Stability	Under constant ambient conditions, total drift for 8 hours following 30 minutes of warmup shall be no greater than 0.15 percent +45 mV.
Operating temperature	0° to 50°C.
Temperature coefficient	(0.05 percent +15 mV)/°C following a 30-minute warmup period.
Output polarity	Positive, negative or floating with respect to ground.
Overload protection	A short circuit load shall not damage the supply.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	Hewlett Packard Harrison Division Model 6439B Power Supply.

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ITEM P05

POWER SUPPLY, CONSTANT CURRENT

TYPE OF EQUIPMENT	Constant current power supply.
FUNCTION PERFORMED	Continuously variable values of constant current within each of three ranges.

SPECIFICATIONS

DC output	
Current	0 to 500 mA.
Voltage	0 to 50 V.
Ranges	0 to 5, 0 to 50, and 0 to 500 mA.
Impedance (R in parallel with C)	0 to 5 mA range: R = 330 MΩ, C = 500 pF 0 to 50 mA range: R = 33 MΩ, C = 0.005 uF 0 to 500 mA range: R = 3.3 MΩ, C = 0.05 uF
The power supply shall have remote programming features.	
Line regulation	25 PPM or less of output ±5 PPM of range setting for line voltage from 115 V ±10 percent.
Constant current load regulation	25 PPM or less of output ±5 PPM of range setting for a load change which causes the output voltage to vary from 0 to 50 Vdc.
Ripple and noise	0 to 5 mA range: 0.40 uArms/3 uA p-p 0 to 50 mA range: 4.0 uArms/40 uA p-p 0 to 500 mA range: 40 uArms/250 uA p-p
Load transient recovery time	200 us or less for output current recovery to within 1 percent of the nominal output current following a full load change in output voltage.
Stability	100 PPM or less of output current ±25 PPM of range setting for eight hours after one hour warmup under conditions of constant load, output setting and temperature.
Operating temperature	0° to 50°C.
Temperature coefficient	Output change/°C is 75 PPM or less of output current ±5 PPM of range setting.
Output polarity	Positive and negative with respect to ground.
Meter	
Ranges	6, 60, 600 mA, 60 Vdc.
Accuracy	2 percent of full scale.

EQUIPMENT MEETING ALL SPECIFICATIONS

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ITEM P06

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A continuously variable dc voltage power source with high output current (60 A) capability.
SPECIFICATIONS	
DC output	0 to 40 Vdc at 0 to 60 A.
Input voltage	187 to 229 Vac or 207 to 253 Vac, 50 to 63 Hz.
Line and load regulation	±0.075 percent or ±15 mV (whichever is greater) with no load to full load and ±10 percent line voltage change combined.
Ripple	0.4 percent + 40 mV.
Stability	0.05 percent of maximum input voltage for 8 hours after 30 minutes warmup.
Temperature coefficient	6 mV/°C.
Operating temperature range	0° to 55°C.
Circuit protection	Continuously adjustable 0 to 60 A current limiting protects load and power supply.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Sorenson Model DCR 40-60A.

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ITEM P07

POWER SUPPLY, DC

TYPE OF EQUIPMENT	DC power supply.
FUNCTION PERFORMED	A continuously variable dc voltage power source for general purpose use in supplying power to integrated circuits (IC).
SPECIFICATIONS	
DC output	0 to 34 Vdc at 0 to 0.5 A.
Input voltage	105 to 125 Vac, 47 to 420 Hz.
Line regulation	0.01 percent or 2 mV (whichever is greater) for 105 to 125 Vac.
Load regulation	0.01 percent or 3 mV (whichever is greater) no load to full load.
Ripple	250 uVrms, 2 mV p-p
Stability	0.05 percent or 10 mV (whichever is greater) for 8 hours after warmup under fixed line load and temperature conditions.
Temperature coefficient	(0.02 percent +400 uV)/°C.
Operating temperature range	0° to 50°C.
Output polarity	Positive, negative or floating with respect to ground.
Circuit protection	Current limiting protects load and power supply.
Remote sensing	Provision is made for remote sensing to eliminate the effect of power output lead resistance on dc regulation.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturers	Trygon Electronics Model DL40-1 Power Designs Model 4010.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Power/Mate Model BP34C.
Exceptions	No provision for remote sensing.
Manufacturer	Power Designs Model 2020.
Exceptions	Output: 20 Vdc.

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ITEM P08

STANDARD, DC VOLTAGE

TYPE OF EQUIPMENT	DC voltage standard, differential voltmeter, and null detector.
FUNCTION PERFORMED	Variable dc power source, differential voltmeter, and null detector.
SPECIFICATIONS	
As a dc voltage standard	
Output voltage	0.0 to 1111.1110 Vdc.
Output current	0.0 to 50 mA.
Voltage ranges	10 V, 100 V, and 1000 V with output of: 1.0 to 11.11110 (1 uV steps) 0.0 to 111.11110 (10 uV steps) 0.0 go 1111.1110 (100 uV steps)
Resolution	0.1 ppm of range (1 uV maximum).
Accuracy of output	10 V range ± (0.002 percent of setting + 10 uV) 100 V range ± (0.002 percent of setting + 20 uV) 1000 V range ± (0.002 percent of setting + 40 uV)
Temperature coefficient of output	Less than (0.002 percent of setting + 1 uV) per °C from 0° to +50°C.
Stability of output	10 V range ± (0.001 percent of setting + 10 uV). 100 V and 1000 V range ± (0.001 percent of setting + 20 uV).
Overcurrent protection	Automatic current limiting, continuously adjustable from 1 to 60 mA with front panel control and indicator. Normal operation is restored upon removal of overload.
Overvoltage protection	Automatic disconnect of load if output voltage exceeds 1 to 1200 V setting on front panel controls. Manual reset.
Ripple and noise	10 V range <20 uVrms. 100 V range <30 uVrms. 1000 V range <40 uVrms.
Setting time	Within 10 ppm of final output, 10 seconds after a range change.
Output resistance	0.0005 Ω or (0.0001 Eo) Ω at dc, whichever is greater.
Regulation for either a 10 percent line voltage change or a full load change	0.0002 percent of setting or 10 uV.

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ITEM P08 - Continued

Common mode rejection (dc to 400 Hz up to 700 Vrms or 1000 Vdc)	Better than 125 dB.
Isolation	Either output terminal may be floated up to 1000 Vdc from chassis ground.
Remote sensing	Separate terminals are provided to sense the output voltage directly at the load, reducing errors caused by voltage drop in connecting wires between the output and load.

As a differential voltmeter

Accuracy	Same as output characteristics listed under DC voltage standard.
Voltage ranges	Same as listed under DC voltage standard.
Stability	Same as listed under DC voltage standard.
Line regulation	0.0002 percent + 10 uV for 10 percent line change from nominal at constant load.
Sensitivity ranges	1000 V to 10 uV (full scale) in nine (9) decade ranges. Any null sensitivity may be used on any voltage range.
Input resistance	Infinite at null from 0.0 to 1111.1110 Vdc.
Meter and dial resolution	0.1 ppm of range.

As a null detector

Accuracy	3 percent of range

Ranges

Voltage range	Input resistance
1000-0-1000	100 MΩ
100-0-100	100 MΩ
10-0-10	100 MΩ
1-0-1	100 MΩ
00.1-0-00.1	10 MΩ
0.01-0-0.01	10 MΩ
0.001-0-0.001	1 MΩ
0.0001-0-0.0001	1 MΩ
0.00001-0-0.00001	1 MΩ

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ITEM P08 - Continued

Stability of meter zero	One most sensitive range (10 uV full scale) 0.5 uV peak-to-peak noise 0.5 uV peak-to-peak stability for 10 percent line voltage variation.
Input power	115/230 Vac ±10 percent, 50 to 60 Hz single phase. Approximately 130 VA fully loaded.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	John Fluke - Model 335A.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Cohu Electronics - Model 351
Exceptions	No differential voltmeter capabilities No null detector capabilities Accuracy of output: 10 V range ±(0.003 percent of setting + 10 uV) 100 V range ±(0.003 percent of setting + 20 uV) 1000 V range ±(0.003 percent of setting + 40 uV)

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ITEM P09

STANDARD, DC VOLTAGE AND NULL METER

TYPE OF EQUIPMENT	DC voltage standard and null meter.
FUNCTIONS PERFORMED	Variable dc voltage power source and null meter.
SPECIFICATIONS	
Output voltage	1,000 to 502.110 V, variable in 100, 10, 1, 0.1, 0.01, and 0.001 V steps.
Output current	Up to 20 mA.
Meter range, full scale	
302 output	±500 V, ±50 V, zero center (see note).
External input	±500 V, ±50 V, zero center (see note).
DC null meter	±50 V, ±5 V, ±0.05 V, zero center (see note).
Stability of output	
One month	±50/0/0m ±50 uV.
Short term (several hours)	±25/0/0m ±25 uV.
Output voltage calibration	0.01 percent of reading, ±0.0002 V.
Output hum and noise	<100 uVrms.
Load regulation	0.002 percent or 200 uV, whichever is greater, no load to full load.
Line regulation	0.002 percent for 10 percent line voltage change.
Temperature coefficient	<10/0/0m/°C from +15° to +35°C.
Meter accuracy	4 percent of full scale.
Source impedance	<(0.01 Ω or 0.001 Eo) ohm, whichever
Floating output	1000 MΩ resistance between chassis and either output terminal; current between either output terminal and chassis is 0.5 uAdc, <4 uA at 60 Hz.
Response time	0.2 ms.
Power requirements	105 to 125 or 210 to 250 V, 50 to 60 Hz, 90 VA.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Cohu/Kintel Model 302.

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ITEM P09 -Continued

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED:

Manufacturer- - - - - - - - - - - - - - - - -	Fluke 335A.
Exceptions- - - - - - - - --- - - - - - - -	Short term stability not specified. Temperature coefficeint- less than (0.002 percent of setting + 1 μV) per °C from 0° to 50°C (DC standard isolation-either output terminal may be floated up to 1000 Vdc from chassis ground, 100 MΩ at less than 70 percent relative humidity, 50°C. Response time (setting time) within 10 ppm of final output, less than 10 seconds after a range change.

NOTE: These ranges are obtained by setting meter range switch to 500 V or 0.5 position and depressing the METER SENS X10 switch.

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ITEM 001

GENERATOR, POWER

TYPE OF EQUIPMENT	Power generator.
FUNCTION PERFORMED	Variable frequency, low distortion, and variable voltage ac power source.
SPECIFICATIONS	
Fixed Frequency	400 Hz.
Accuracy	The stabilized fixed frequency is factory adjusted to ±0.1 percent of the nominal value at room ambient of 25°C.
Stability	After warmup, fixed frequency should not change by more than ±0.2 percent (includes normal temperature variation and tube aging).
Distortion	Less than 0.5 percent.
Regulation	
Load	Less than ±0.5 percent no load to full load.
Line	Less than ±0.5 percent at output for line variations of ±5 percent, at full power output.
Output power	160 VA.
Output voltage	0 to 130 V, isolated output.
Variable frequency	350 to 450 Hz.
Power Amplifier (External source)	45 to 5,000 Hz. (Usable to 10 kHz at reduced power output).
Hum level	Approximately 70 dB below rated output voltage.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Industrial Test Equipment Model *160GN.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Tel-Instrument Electronics Model 4025B-1-E12A Model 4025B-1-J2A.
Exceptions	The 4025B-1-E12A meets the adjustable 350-450 Hz specifications while the 4025B-1-J2A meets the fixed frequency requirements. Broadband operation with external oscillator inputs on all units 50 Hz to 4 kHz.

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ITEM R01

OHMMETER
(Obsolescent)

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Ohmmeter.

FUNCTION PERFORMED - - - - - - - - - - - - - Measures resistance over a wide range.

SPECIFICATIONS

Resistance range - - - - - - - - - - - - - From 1Ω center scale to 100 MΩ center scale, in decade steps.

Accuracy - - - - - - - - - - - - - - - - - - ±5 percent reading from 0.2Ω to 500 MΩ; ±10 percent reading from 0.1 to 0.2Ω, and from 500 to 5,000 MΩ.

Voltages and currents - - - - - - - - - - -

Range (ohms)	Open circuit volts (dc)	Short circuit current (dc)
x1	10 mV	10 mA
x10	100 mV	10 mA
x100	1 V	10 mA
x1000	1 V	1 mA
x10 k	1 V	100 uA
x100 k	1 V	10 uA
x1 M	1 V	1.0 uA
x10 M	1 V	0.1 uA
x100 M	1 V	0.01 uA

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard Model 412A.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - - Conu Model 208R.

Exceptions - - - - - - - - - - - - - - - - Terminal voltage is 1 V full scale all ranges.

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ITEM R02

BRIDGE, RESISTANCE

TYPE OF EQUIPMENT - - - - - - - - - - - - - - - Resistance bridge.

FUNCTION PERFORMED - - - - - - - - - - - - - - Measures dc resistance and conductance of 2-terminal passive networks.

SPECIFICATIONS

Resistance

Range - - - - - - - - - - - - - - - - - - - 0 to 1,200 kΩ in 7 ranges, 1 step per decade. (0.01 MΩ per dial division on lowest range).

NOTE: For resistance measurement capability below 1Ω, bridge requires special wiring.

Accuracy - - - - - - - - - - - - - - - - - High-and-low ranges; 0.1 percent ± 1 dial division. Other 5 ranges; 0.05 percent ±1 dial division.

Conductance

Range - - - - - - - - - - - - - - - - - - - 0 to 1,200 mmhos in 7 ranges. (10 μmhos per dial division on lowest range).

Accuracy - - - - - - - - - - - - - - - - - High-and-low ranges; 0.1 percent ± 1 dial division. Other 5 ranges; 0.05 percent ± 1 dial division.

Source/detector

A dc generator shall be supplied as an integral part of the bridge or as a separate unit of such design that the bridge cannot be damaged by improper adjustment of controls. If interconnection is required between source and bridge, proper attention should be given to prevent accidental damage. The generator voltage output and impedance shall be such as to realize full accuracy, sensitivity, and utility of the bridge.

A detector shall be provided internal or external to the bridge for sensing nulls. The detectors utilized shall be such that the sensitivity and accuracy of the bridge are fully realized.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Electro Scientific Industries Model *290A Impedance Bridge used in conjunction with an *840 (or *803) dc generator-detector.

EQUIPMENT MEETING SPECIFICATION
WITH EXCEPTIONS AS NOTED

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ITEM R03

BRIDGE, RESISTANCE, RF

TYPE OF EQUIPMENT	RF resistance bridge.
FUNCTION PERFORMED	Measures the equivalent parallel resistance of a 2-terminal circuit at radio frequencies.
SPECIFICATIONS	
Resistance	
Range	15 to 100,000Ω.
Dial increments	3 percent nominal.
Accuracy	$\pm(2+0.5\times10^{-8}F+0.1\ \Omega FC)\%$ $\pm0.2\Omega$ where C is equivalent 2-terminal capacitance (parallel) at measurement frequency.
Frequency generator	
Range	500 kHz to 250 MHz in 9 ranges (octave steps, approximately)
Accuracy (dial)	±1 percent
Increments (dial)	1 percent nominal.
Detector	Tuned, null indicator.
DC current capacity	50 mA maximum through input terminals.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard HP 250A.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM R04

OHMMETER

(Obsolescent)

TYPE OF EQUIPMENT	Ohmmeter.
FUNCTION PERFORMED	Accurate measurement of resistance.
SPECIFICATIONS	
Range	100Ω to 10 MΩ full scale in 6 decades (1 step/decade) with a minimum of 300 percent overranging.
Measurement current	1/R full scale amperes except 10 MΩ range where 1-1 uA.
Accuracy, overall	±(0.01 percent full scale) ±(0.02 percent of reading ±1 digit) ±absolute value of (0.001 percent full scale ±0.003 percent of reading) per °C change from 25°C when calibrated daily against internal standard.
Internal standard	±1 volt dc for self-calibration; stability is ±0.005 percent per 6 months maximum at +25°C; temperature coefficient ±0.001 percent per °C (+10° to +40°C).
	NOTE: 1 Vdc is the full scale output for the ohms to dc voltage converter on all ranges. It is also a full scale output of the dc to digital converter (DVM).
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Dymec Division of Hewlett-Packard . Model *DY 2410B ac volts/ohms converter with DY 2401C DVM.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

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ITEM R05

OHMMETER

(Obsolescent)

TYPE OF EQUIPMENT	Ohmmeter.
FUNCTION PERFORMED	Accurate measurement of resistance.
SPECIFICATIONS	
Range	100Ω to 10 MΩ full scale in 6 decades (1 step/decade) with a minimum of 300 percent overranging.
Measurement current	1/R full scale amperes except 10 MΩ range where I=1 μA.
Accuracy, overall	(±0.01 percent reading ±0.01 percent full scale ±1 digit) ±0.004 percent reading ±0.0013 percent full scale/°C change from 25°C when calibrated daily at 25°C against internal standard.
	NOTE: When overranging, percent of full scale errors are converted to percent of reading and added to percent of reading errors.
Internal Standard	±1 Vdc for self-calibration; stability ±0.01 percent per month maximum at +25°C; temperature coefficient ±0.002%/°C (+10° to +40°C).
	NOTE: 1 Vdc is the full scale output for the ohms to dc voltage converter on all ranges. It is also a full scale output of the dc to digital converter (DVM).
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Dymec Division of Hewlett-Packard Model *DY 2410A with 2401A DVM.
Item	R04.

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ITEM R06

BRIDGE, MEGOHM

TYPE OF EQUIPMENT	High-resistance bridge.
FUNCTION PERFORMED	Measures (a) insulation resistance, resistance films, high valued resistors, semiconductors, capacitor dissipation; (b) voltage and temperature coefficients of resistance.

SPECIFICATIONS

Resistance	
Range	1 kΩ to 10^{15}Ω in 10 ranges.
Accuracy	1 kΩ to 10^{10}Ω ±1 percent; after self-calibration: 10^{10}Ω to 10^{12}Ω ±1 percent at high voltages; 1 percent accuracy is obtainable at 10 V up to 10^{11}Ω; 10^{13}Ω ±2 percent; 10^{14}Ω ±10 percent; 10^{15}Ω ±1 scale division.
Delta R percent dial	±5 percent range; accurate to ±0.2 percent.
Test voltage	
Fixed voltage accuracy	±3 percent to ±0.5 V.
Fixed voltages and corresponding minimum unknown resistances	10 V-1 kΩ; 20 V-3 kΩ; 50 V-7 kΩ; 100 V-20 kΩ; 200 V-50 kΩ; 500 V-150 kΩ; 1 kV-500 kΩ.
Variable voltage	Adjustable with external potentiometer.
Minimum test voltage for 1 percent resolution	10 V with 100 G or less multiplier for maximum unknown R=10^{11}Ω; 100 V with 100 G multiplier for maximum unknown R=10^{12}Ω; 200 V with 1 T multiplier for maximum unknown R=10^{13}Ω.
Short circuit current	<15 mA, 10 to 15 V; <10 mA, 100 to 1,000 V.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer	General Radio Co. Type 1644-A, megohm bridge.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

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ITEM R07

OHMMETER

TYPE OF EQUIPMENT	Ohmmeter.
FUNCTION PERFORMED	Accurate measurement of resistance.
SPECIFICATIONS	
Range	Resistance measurements from 0.1Ω to 10 MΩ with the following ranges: 1.0 kΩ full scale +20 percent over-range 10.0 kΩ full scale +20 percent over-range 100.0 kΩ full scale +20 percent over-range 1,000.0 kΩ full scale +20 percent over-range 10,000.0 kΩ full scale
Measurement current	10/R full scale amperes for all ranges except the 1 kΩ range which is 1/R full scale amperes.
Accuracy	
1 to 100 kΩ	±(0.05 percent of reading +0.02 percent full scale)
1 MΩ	±(0.10 percent of reading +0.03 percent full scale)
10 MΩ	±(0.20 percent of reading +0.05 percent full scale)
Temperature coefficient	
1 kΩ to 1 MΩ	+0.005 percent of reading ±0.0015 percent of full scale/°C.
10 MΩ	+0.005 percent of reading +0.005 percent of full scale/°C.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Fairchild Model 7000 Digital multimeter with option 03 plug-in board.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Fairchild Model 7100A digital voltmeter
Exception	Does not have 1.0 kΩ full scale range.

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ITEM R06

ATTENUATOR

TYPE OF EQUIPMENT	Attenuator.
FUNCTION PERFORMED	Provides attenuation from 0 to 110 dB in 1 dB steps.
SPECIFICATIONS	
Accuracy	(a) 1 dB steps; dc to 100 kHz, error ±0.125 dB/step; from 100 kHz to 1 MHz, ±0.25 dB/step. (b) 10 dB steps; from dc to 100 kHz, error ±0.25 dB in steps up to 70 dB, ±0.5 dB/step above 70 dB: from 100 kHz to 1 MHz, error ±0.5 dB at any step up to 70 dB; ±0.75 dB/step above 70 to 100 dB.
Frequency	DC to 1 MHz.
Power capacity	5 W (55 Vdc or rms), continuous duty.
Input-output impedance	600Ω.
DC isolation	Signal ground level from external chassis ±500 Vdc minimum.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 350/D.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM R09

ATTENUATOR, BALANCED

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Balanced attenuator.

FUNCTIONS PERFORMED - - - - - - - - - - - - Provides voltage attenuator form 0 to 120 dB in 0.1 dB steps. Provides equal resistance into each terminal.

SPECIFICATIONS

Accuracy - - - - - - - - - - - - - - - - -

Attenuation	100 kHz	1 MHz	1.5 MHz *
0 to 60 dB	±0.1 dB	±0.2 dB	±0.2 dB
60 to 90 dB	±0.1 dB	±0.3 dB	±0.3 dB
90 to 110 dB	±0.2 dB	±0.5 dB	±0.5 dB
110 to 119.9 dB	±0.3 dB	±1.0 dB	

*Typical value

Input/output impedance - - - - - - - - - - 600 Ω ± 10 Ω.

Unbalance capacitance - - - - - - - - - - <10 pF.

Maximum input power - - - - - - - - - - - 1 W.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard Model 4436A.

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ITEM R10

OHMMETER

TYPE OF EQUIPMENT	Digital multimeter.
FUNCTIONS PERFORMED	Measures AC, DC current and resistance. (The low short circuit current (1 uA) at the 200 kΩ range is the most desirable function).
SPECIFICATIONS	
DC current	
Ranges	±199.9 uA to 10 A in five increments (1 minute operation from 10 to 20 A).
Accuracy	±0.3 percent of reading +1 digit, all ranges except 10 A.
Voltage burden	0.5 V max up to 10 A.
Response time	0.5 second.
Maximum input	2 Adc (fuse protected) separate input to 20 A (not fused).
Operating time	10 A and below, continuous. Above 10 A, one minute maximum (duty cycle 25 percent).
AC current	
Values are the same as the DC current for Range, Voltage burden, Max input, and Operating time.	
Accuracy	45 Hz to 10 kHz, ±1.0 percent of reading + 2 digits.
Response time	3 seconds worst case.
Resistance	
Ranges	199.9Ω, 1.999 kΩ, 19.99 kΩ, 199.9 kΩ, 1999 kΩ, and 19.99 MΩ.
Accuracy	±0.2 percent of reading + 1 digit on all ranges except: ±0.5 percent of reading + 1 digit on 20 MΩ range.
Response time	1/2 second on all ranges except 4 seconds on 20 MΩ range.
Current through unknown	200 Ω and 2 kΩ = 1 mA 20 kΩ = 100 uA 200 kΩ and 2000 kΩ = 1 uA 200 MΩ = 0.1 uA
Maximum input	200 Ω and 2 kΩ ranges, 130 Vrms.
Voltage	20 kΩ through 2 MΩ ranges, 250 Vrms (Note display flashes 1999 with open test leads telling operator not to apply to voltage source without changing function).

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ITEM S01

WATTMETER, AC

TYPE OF EQUIPMENT	AC wattmeter.
FUNCTION PERFORMED	Measures single phase, two phase and three phase ac power.

SPECIFICATIONS

Wattage	
Range	0.1 to 120 W.
Accuracy	±3 percent of reading.
Frequency range	50 to 2,000 Hz.
Waveform error	<2 percent for 5 percent harmonic distortion.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	Voltron Model PW-3.
Exceptions	Range To 30 W at 26 V. To 120 W at 115 V. Power factor 0.2 to 1.0 lag or lead at 50<F<200 Hz. 0.1 to 1.0 lag or lead at 200<F<2,000 Hz. Accuracy 1 percent of full scale.

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ITEM TDI

COUNTER - TIMER

TYPE OF EQUIPMENT	Counter-timer.
FUNCTION PERFORMED	The counting of separate electrical events capable of being referred to a known time period to display events in digital form; i.e., frequency, period, and ratios of frequencies.
SPECIFICATIONS	
Frequency	
Range	0 to 13 MHz.
Accurancy	±1 count ± time base accuracy.
Gate time	1 us to 10 s in decade steps.
Period measurements	
Range	
Single period	0 to 3 MHz.
Multiple period	0 to 1 MHz.
Multiple period	1 to 10^7 periods in decade steps.
Accuracy	±1 count ± time base accuracy ± trigger error (trigger error less than 1 us with a 1 kHz sine wave).
Frequency counter	
Single period	10^7 to 0.1 Hz.
Multiple period	10^6 or 10^7 Hz.
Trigger temperature stability	2 cyclic degrees /°C maximum for 100 mVrms input.
Ratio measurements	
Display	(B/A) times multiple period.
Range	(a) 0 to 1 MHz, dc coupled. (b) 1 Hz to 13 MHz. NOTE: 6 dB/Octave + rolloff below 10 Hz, ac coupled.
Sensitivity (A and B)	100 mVrms, 1 MΩ input resistance shunted by 30 pF.
Accuracy	±1 count of B ± trigger error of A; where B is frequency applied to counting binaries and A is applied to time base decade dividers.
Time interval measurement (A and B)	
Range	0.3 us to 10^8 s.
Accuracy	±1 count ± trigger level error each channel ± oscillator accuracy.
Frequency counted	0.1 Hz to 10 MHz in decade steps.
Start and stop	Independent channels.
Time interval with external standard (C input, (C/A→B)	
Range	10 Hz to upper limit of frequency measurement range. Time A→B ± 0.1 us. (Start A, stop B).

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Accuracy	±1 count ± accuracy of input to clock C ± trigger level error.
C sensitivity	1 Vrms, 2,500Ω input.
Count A	1 to 10^8 impulses.
Range	1 to 10^8 impulses.
Frequency response	0 to 13 MHz.
Time base	
Frequency	1 MHz.
Oscillator stability	±5 parts in 10^8 over 3 hour period (after 3-hour warmup and at constant ambient temperature); ±2 parts in 10^7 after 15 minute warmup; ±5 parts in 10^7 per month long term after ultimate stability is reached.
Output frequencies	0.1 Hz to 1 MHz in decade steps; and 1 MHz and 10 MHz unselectable.
Display	6 digit readout.
Sample ratio	Gate time is fixed but hold time is variable from 0.1 to 15 s. Display may be held indefinitely.
Input	
A and B rated sensitivity	100 mVrms (entire frequency range).
Basic sensitivity	Approximately 10 mVrms below 15 MHz. This is the amplitude needed to just allow the instrument to count, but may not give a stable reading.
Maximum C sensitivity	1 Vrms.
Coupling	(a) A - ac or dc. (b) B - ac or dc. (c) C - ac.
External Frequency Standard	1 MHz, 1 Vrms into 1 kΩ.
Attneuation	
A and B amplifiers	Step attenuator 0.4, 1.6, 6.4, 26, 102, T. (The latter levels indicate approximate amplifier saturation level in volts p-p). T indicates input of amplifier terminated to ground, not grounded.
Impedance	1 MΩ resistance shunted by 30 pF. No overload protection. Maximum input A and B amplifiers in 0.4 attenuator range 300 Vp-p, 300 v peak in all other attenuator positions.
Outputs	(a) 1 MHz (nominal) 5 Vp-p from a 4.7 kΩ source; (b) 10 MHz (nominal) 8 Vp-p from a 1 kΩ source.
Operating temperature range	0° to 55°C.

EQUIPMENT MEETING
ALL SPECIFICATIONS

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ITEM T02

FREQUENCY METER

(Obsolescent)

TYPE OF EQUIPMENT	Frequency meter.
FUNCTION PERFORMED	Frequency measurement.
SPECIFICATIONS	
Range	5 Hz to 300 kHz
Accuracy	±1 count ± time base error.
Stability, time base	2 parts/10^6/week, ±2.5 parts/10^6 average (+10°C to +50°C).
Input circuit	Resistance: 1 MΩ. Capacitance: 110 pF.
Display	6 digits (5 digits plus 1 digit for over-ranging), decimal point, and units.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Dymec Division of Hewlett-Packard Model 2401A Model 2401C.
Item	T01

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ITEM T03

FREQUENCY METER

(Obsolescent)

TYPE OF EQUIPMENT	Frequency meter.
FUNCTION PERFORMED	Measurement of sinusoid and pulse train frequencies from 5 Hz to 1,000 kHz.
SPECIFICATIONS	
Range	5 Hz to 1,000 kHz.
Accuracy	$\pm$0.005 percent of reading.
Stability	$\pm 5 \times 10^{-6}$ per month.
Sensitivity	500 mVp-p.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Wavetek Model 210 with Model 2103 Hertz/DC Volt/Ratio plug-in assembly.
Exceptions	Accuracy - $\pm$0.005 percent of reading, $\pm$0.001 percent of range at constant temperature. Special features - This instrument can also be used for measuring dc voltage and dc voltage ratios. With the plug-in unit Model 2102, ac voltages can also be measured.

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ITEM 001

LEAKAGE TESTER, HIGH VOLTAGE DC

TYPE OF EQUIPMENT	High voltage dc leakage tester.
FUNCTION PERFORMED	Provide high voltage dc insulation breakdown/leakage tests.
SPECIFICATIONS	
Range	
Voltage	0 to 2,500 Vdc continuously variable.
Leakage meter (minimum)	0 to 0.5 mA.
Useful megohm	1 to 50 MΩ.
Maximum short circuit current	5 mA.
Indicators	
Voltmeter	An internally wired meter with range as specified shall be provided.
Visual	A visual indicator shall be provided to indicate when the test leads are energized.
Audible	A speaker will provide an audible means of detecting arcing and breakdown.
Safety provisions	
Power	Power to test leads shall be applied via a momentary type (spring loaded) switch.
Test leads	Test leads and clips shall be covered with suitable insulation to prevent shorting.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Slaughter Company Model 105-2.5.
EQUIPMENT MEETING SPECIFICATION WITH EXCEPTIONS AS NOTED	
Manufacturer	Slaughter Company Model 103/105-2.5.
Exceptions	Maximum short-circuit current 10 mA.

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ITEM 002

LEAKAGE TESTER, HIGH VOLTAGE AC

TYPE OF EQUIPMENT	High voltage ac leakage tester.
FUNCTION PERFORMED	Provide high voltage ac insulation breakdown/leakage tests.
SPECIFICATIONS	
Range	
Voltage	0 to 2,500 Vac continuously variable.
Leakage meter (minimum)	0 to 5 mA.
Useful megohm	0.1 to 5 MΩ.
Capacitance compensation	0 to 0.005 uF.
Maximum short-circuit current	10 mA.
Indicators	
Voltmeter	An internally wired meter with range as specified shall be provided.
Visual	A visual indicator shall be provided to indicate when the test leads are energized.
Audible	A speaker will provide an audible means of detecting arcing and breakdown.
Safety provisions	
Power	Power to the test leads shall be applied via a momentary type (spring loaded) switch. Test leads and clips shall be covered with a suitable insulation to prevent shorting.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Slaughter Company Model 103-2.5. Slaughter Company Model 103/105-2.5.

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ITEM UQJ

TESTER, LSI

TYPE OF EQUIPMENT	Large scale integration tester.
FUNCTION PERFORMED	The LSI tester is comprised of a parallel programmed, special-purpose multiprocessor and a personality board. The multiprocessor is a microprogrammable system which utilizes programmable test routines that are run continuously at real time speeds. The multiprocessor is capable of testing devices/systems whose operations can be simulated or tested by algorithm. The multiprocessor supplies addressing, data, and control information to the personality board and responds to data and control information returning from the personality board. The personality board, when connected with the multiprocessor I/O port, produces the necessary voltage levels and timing required to test a device/system.
SPECIFICATIONS	
Multiprocessor description	Display pipeline for correct error information display. Microinstruction loading by paper tape reader or manual interface. Fifteen levels of subroutine and indexing. All inputs and outputs are TTL compatable except where noted. Programmed testing rates of 3 Hz, 5 MHz, and 10 MHz; 0 to 10 MHz from external clock input (panel). TTL positive going programmable sync. output (panel). TTL positive going error signal output (panel). Machine clock output (panel). Single step, looping, error override and external testing modes. Dual 80 pin input/output port. 115 Vac ±10 percent at 3 A power requirement.
Multiprocessor outputs (to the personality board)	16 address bits, A0 to A15 and $\overline{A0}$ to $\overline{A15}$. 16 data bits, T0 to T15 and $\overline{T0}$ to $\overline{T15}$. 8 interface sequence bits, Y0 to Y7. Multiprocessor clock, CLK. Cycle initiate command, CIN and $\overline{CIN}$. Read cycle command, RCY and $\overline{RCY}$. Write cycle command, WTC and $\overline{WTC}$. Cycle pulse, $\overline{CIP}$. Read pulse, $\overline{RCP}$. Write pulse, WIP. Program terminate pulse, PTP. Program pass indicator, PAS. Program fail indicator, FAL. External data strobe, DSX (from panel). External write pulse, EWP (from panel). 5 V ±1 percent at 1 A nonadjustable power supply. Two ±3 to ±25 V at 0.5 A resistor programmable power supplies.

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Multiprocessor inputs (from the personality board)	16 data bits, D0 to D15. Personality card generated clock, EXC. Clock frequency selector, CLF. Multiprocessor clock halt, HLT. Data strobe input, DSI. Buffer memory write pulse, WPE. Personality board start input, SRT. Program interrupt, INT. Flag signal, FLG.
Personality board description	A programmable V_{cc} level. Logic level thresholds for inputs to device/system. Programmable timing delays to the device/system for addressing, input data, enabling, writing, and reading, at the specified voltage level, of data from the device under test. Device/system cycle rate.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer	Macrodata MD-104 with personality board.

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ITEM V01

VOLTMETER, DC
(Obsolescent)

TYPE OF EQUIPMENT	DC voltmeter.
FUNCTION PERFORMED	Accurately measures dc voltages over wide ranges.
SPECIFICATIONS	
Voltage range	1 mVdc full scale to 1,000 Vdc full scale (2 steps/decade 1, 3 sequence).
Accuracy	±1 percent full scale on any range.
Input resistance	10 MΩ ±1 percent on 1/3/10 mV ranges 30 MΩ ±1 percent on 30 mV ranges 100 MΩ ±1 percent on 100 mV ranges 200 MΩ ±1 percent on 300 mV range and above.
AC rejection	A voltage at power line or twice power line frequency 40 dB greater than full scale affects reading less than 1 percent. Peak voltage must not exceed 1,500 v.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 412A.
Manufacturer	Conu Model 208R.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	

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ITEM V02

VOLTMETER, DC

(Obsolescent)

TYPE OF EQUIPMENT	Digital dc voltmeter (integrating type).
FUNCTION PERFORMED	DC voltage measurement.
SPECIFICATIONS	
Range	0.1 to 1,000 Vdc full scale in 5 decades with minimum over-ranging of 300 percent without overload (no over-ranging on 1,000 V range).
Accuracy (at +25°C)	±0.01 percent reading ±0.005 percent full scale ±1 digit (0 to full scale) ±0.02 percent reading ±1 digit (at 2 x full scale) ±0.025 percent reading ±1 digit (at 3 x full scale)
	NOTE: Specifications hold for all ranges ±10 percent line voltage change and 6 months operation. Assume daily calibration against internal standards at +25°C.
Temperature coefficient	
When calibrated against internal standard at operating temperature	±0.001 percent reading/°C (+10° to +40°C) ±0.0015 percent reading/°C (+40° to +50°C).
When not calibrated at operating temperature, over the temperature range +10° to +50°C	±0.002 percent reading ±0.0005 percent full scale/°C (0.1 V range). ±0.002 percent reading ±0.0002 percent full scale/°C (1/10/100/1,000 V ranges).
Internal calibration source	±1 Vdc internal standard provided for self-calibration. May be compared to external standard and adjusted to within ±0.001 percent. Factory adjusted to within ±0.002 percent at +25°C.
Temperature coefficient	±0.001 percent/°C (+10° to +40°C); and ±0.0015 percent/°C (+40° to +50°C).
Input	
Isolation	Floated and guarded signal pair; signal pair and guard may be operated up to 500 V above chassis ground.
Resistance	100 kΩ on 0.1 V range 1 MΩ on 1 V range 10 MΩ on 10/100/1,000 V ranges.
Capacitance	150 pF maximum all ranges.
Sample periods	0.01, 0.1, 1.0 s integration time.
Spurious signal rejection	
Common mode	140 dB at all frequencies for 0.1 s sample period; 160 dB at dc frequencies for 0.1 s sample period; 120 dB at 60 Hz for 1,000 in common lead; 160 dB at dc for 1,000 in common lead.

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V02 - Continued

Noncommon mode - - - - - - - - - - - - - -	20 dB minimum at 55 Hz for 0.1 s sample period; increases 20 dB per decade with increase in frequency.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - -	Dymec Division of Hewlett-Packard Model DY 2401C (digital dc voltmeter) used in conjunction with Model DY 2401B (ac/ohms converter).

EQUIPMENT MEETING SPECIFICATION
WITH EXCEPTIONS AS NOTED

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ITEM V03

VOLTMETER, DC

(Obsolescent)

TYPE OF EQUIPMENT	Digital dc voltmeter (integrating type).
FUNCTION PERFORMED	DC voltage measurement.
SPECIFICATIONS	
Range	0.1 to 1,000 Vdc full scale in 5 decades with minimum over-ranging of 300 percent without overload (no over-ranging on 1,000 V range).
Accuracy	
Stability (at constant temperature)	±0.01 percent of full scale/day 1/10/100/1,000 V ranges. ±0.02 percent of full scale/week 1/10/100/1.000 V ranges. ±0.03 percent of full scale/day 0.1 V range ±0.06 percent of full scale/week 0.1 V range.
Linearity (terminal)	±0.005 percent of full scale, zero to full scale linearity decreases to ±0.015 percent of reading at 200 percent of full scale; and ±0.075 percent of reading at 300 percent of full scale.
Attenuator	Factory-calibrated to ±0.005 percent of 1 V range.
Least significiant digit	A ±1 digit ambiguity exists at all times; this error can be minimized by operating at maximum sample time.
Temperature effects	
Scale factor	±0.002 percent of reading/°C, 0.1/1 V ranges; ±0.0027 percent of reading/°C, 10/100/1,000 V ranges.
Zero (calibratable)	±0.002 percent of full scale/°C, 0.1 V range; ±0.0008 percent of full scale/°C, 1/10/100, 1,000 V ranges.
Internal calibration source	±1 Vdc internal standard provided for self-calibration; may be compared to external standard and adjusted to within ±0.001 percent. factory adjusted to within ±0.002 percent at +25°C.
Resolution	Maximum of 1 uV on the 0.1 V full scale range increasing in decade steps to 0.01 V on the highest range.
Input	
Isolation	Floated and guarded signal pair. Signal and guard may be operated up to 500 V above chassis ground.
Resistance	100 kΩ on 0.1 V range and 1 MΩ on all other ranges.
Capacitance	150 pF maximum.
Sampling periods	0.01, 0.1, 1.0 s integration time.

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Spurious signal rejection	
Common mode	140 dB at all frequencies and 160 dB at dc for 0.1 s sample period; 120 dB at 60 Hz for 1,000Ω in common lead; 160 dB at dc for 1,000Ω in common lead.
Noncommon mode	20 dB minimum at 55 Hz for 0.1 s sample period; increases 20 dB per decade with incease in frequency.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer	Dymec Division of Hewlett-Packard Model DY 2401A (digital dc voltmeter) used in conjunction with Model DY2410A (ac/ohms converter).

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Items V02	
Exceptions	Input resistance 10 MΩ at 10/100/1,000 V full scale.

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ITEM V04

VOLTMETER, DC

TYPE OF EQUIPMENT	Digital dc voltmeter.
FUNCTION PERFORMED	DC voltage measurement.
SPECIFICATIONS	
Range	1.0000 V full scale +20 percent over-range 10.000 V full scale +20 percent over-range 100.00 V full scale +20 percent over-range 1,000.0 V full scale
Accuracy	±0.01 percent of reading ±0.01 percent of full scale; above 500 V ±0.01 percent of reading ±0.027 percent of full scale.
Resolution	100 μV (1 V range).
Input impedance	1 V: > 1,000 MΩ 10 V: > 1,000 MΩ 100 V: > 10 MΩ 1,000 V: > 10 MΩ.
Temperature coefficient	±0.002 percent of reading ±0.008 percent of full scale/°C.
Voltage coefficient	±0.00005 percent/V if input exceeds 500 V.
Temperature range	+5° to +45°C.
Normal mode rejection	>30 dB at 60 Hz >40 dB at 120 Hz >50 dB at 400 Hz
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Fairchild Model 7000 Digital Multimeter.

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ITEM V05

VOLTMETER, DC

TYPE OF EQUIPMENT	DC microvoltmeter
FUNCTION PERFORMED	Measurement of low level dc voltage.
SPECIFICATIONS	
Accuracy	±5 percent of reading or better.
Range	1 μV full scale to 1 V full scale.
Input resistance	Greater than 1 MΩ.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Keithley Instruments Model 150A microvoltammeter
Exceptions	Accuracy - 2 percent of full scale (Worst case accuracy occurs at 3/10 of full scale. Therefore, on a percent of reading basis, the instrument accuracy is ±.67 percent of reading or better). Special features - This instrument can also be used as follows: DC ammeter (Item X03). DC amplifier. Null detector.

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ITEM V06

VOLTMETER, DC

(Obsolescent)

TYPE OF EQUIPMENT	DC voltmeter.
FUNCTION PERFORMED	Measurement of dc voltages from 1 to 1,000 V full scale.
SPECIFICATIONS	
Accuracy	±0.005 percent of reading.
Range	1 to 1,000 V full scale.
Input resistance	1 MΩ, or greater.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Wavetek Model 210 with dc plug-in Model 2101.
Exceptions	Accuracy ±0.005 percent of reading, ±0.00005 percent of range ±5 μV at 25° ±5°C.
	Special features - Six-digit readout/null balance instrument. With the proper plug-in units, this instrument can measure ac voltage and frequency.

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ITEM Y07

VOLTMETER, DIFFERENTIAL, AC/DC

TYPE OF EQUIPMENT	AC/DC differential voltmeter.
FUNCTION PERFORMED	AC/DC voltage measurement
SPECIFICATIONS	
Accuracy	
DC at nominal calibration temperature of 23° ±1°C	±(0.0025 percent of input +0.0001 percent of range +5 uV).
AC at nominal calibration temperature of 23° ±1°C	(a) 30 Hz to 5 kHz; ±(0.05 percent of input +0.0025 percent of range) from 0.001 to 500 Vac and ± 0.1 percent of input from 500 to 1,100 Vac. (b) 5 to 10 kHz; ±(0.07 percent of input +0.0005 percent of range) from 0.001 to 500 Vac and ±0.1 percent of input from 500 to 1,100 Vac. (c) 10 to 20 kHz; ±(0.15 percent of input +0.01 percent of range) from 0.001 to 1,100 Vac.
Range, ac and dc	Input range/null range 1 /0.1, 0.01, 0.001, 0.0001 V 10 /1, 0.1, 0.01, 0.001, 0.0001 V 100 /10, 1, 0.1, 0.01, 0.001 V 1,000/100, 10, 1, 0.1, 0.01 V +10 percent over-range on all ranges.
Resolution, ac and dc	1 V : 1 uV 10 V : 10 uV 100 V : 100 uV 1,000 V : 1 mV.
Input resistance	
DC	1 V : > 1,000 MΩ 10 V : > 1,000 MΩ 100 V : > 10 MΩ 1,000 V : > 10 MΩ
AC	1 MΩ, 40 pF.
Input polarity	Positive and negative.
Temperature range	
DC	16° to 32°C (60° to 90°F): with accuracy of ±(0.005 percent of input +5 uV).
AC	16° to 32°C (60° to 90°F): (a) 20 Hz to 5 kHz ±(0.1 percent of input +25 uV) from 0.001 to 1,100 Vac; (b) 5 to 10 kHz ±(0.15 percent of input +25 uV) from 0.001 to 1,100 Vac; (c) 10 to 20 kHz ±0.3 percent of input from 0.1 to 1,100 Vac;

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V07 - Continued

(d) 20 to 50 kHz ±0.5 percent of input from 0.1 to 110 Vac;
(e) 50 to 100 kHz ±1 percent of input from 0.1 to 110 Vac;
(f) 10 to 20 Hz ±(0.3 percent of input +100 uV) from 0.001 to 1,100 Vac;
(g) 5 to 10 Hz ±(1 percent of input +250 uV) from 0.001 to 1,100 Vac.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - John Fluke Mfg. Co., Inc., any one of the following models:
(1) 887A
(2) 887AB

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - - John Fluke Mfg. Co., Inc., any one of the following models:
(1) 885A
(2) 885AB

Exceptions - - - - - - - - - - - - - - - - (a) Measures dc only.

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ITEM W01

VOLTMETER, AC

TYPE OF EQUIPMENT	Phase angle voltmeter (PAVM), total mode.
FUNCTION PERFORMED	Measurement of voltages over a wide frequency range. (Indicated rms voltage is proportional to average value of applied sine wave.)
SPECIFICATIONS	
Voltage range	1 mV full scale to 300 V full scale in 12 ranges.
Frequency range	10 Hz to 50 kHz direct (useful frequency range with input transformer 50 Hz to 10 kHz).
Accuracy	±2 percent of full scale, direct (10 Hz to 50 kHz).
Calibration	Zero center, low friction meter calibrated with 3-0-3 and 10-0-10 scales.
Input impedance	
Direct mode	10 MΩ, nominal.
Transformer mode	300 kΩ at 400 Hz; 1 MΩ at resonance, nominal.
Input voltage levels	
DC volts	
Direct mode	400 Vdc maximum.
Transformer mode	0 V.
AC volts	
Direct mode	300 Vrms maximum.
Transformer mode	1.0 x F or 300 V, whichever is less. All dc components must be blocked.
Noise	Less than 5 μV for low impedance inputs.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer	North Atlantic	Model *VM204 series Model 214 series Model VM202 series Model VM301 series.

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ITEM W02

VOLTMETER, AC

(Obsolescent)

TYPE OF EQUIPMENT	AC digital voltmeter.
FUNCTION PERFORMED	Measurement of ac voltages (indicated rms voltage is proportional to average value of applied sine wave).
SPECIFICATIONS	
Voltage range	0.1 to 1,000 Vrms full scale in 5 ranges with 300 percent over-ranging without overload except on 1,000 V range which is usable to 750 v peak.
Frequency range	50 Hz to 100 kHz.
Accuracy	(For 3 months when calibrated daily with internal standard) for 20 Hz: ±0.25 percent reading ±0.05 percent full scale; for 30 Hz: ±0.15 percent reading ±0.05 percent full scale; for 50 Hz to 10 kHz: ±0.10 percent reading ±0.05 percent full scale; for 10 to 30 kHz: ±0.20 percent reading ±0.06 percent full scale; for 30 to 100 kHz: ±0.30 percent reading ±0.10 percent full scale. NOTE: 1. During over-ranging, full scale error is to be ignored and percent reading error is to be doubled. 2. ±1 count display ambiguity plus possible error due to response time or output ripple are to be added to above listed data on accuracy.
Temperature coefficient	Over range +10° to +50°C about +25°C for 50 Hz to 10 kHz: ±0.005 percent reading/°C; for 10 to 30 kHz: ±0.01 percent reading/°C; for 30 to 100 kHz: ±0.15 percent reading/°C. NOTE: Calibration at operating temperature is not assumed.
Input impedance	1 MΩ shunted by 100 pF.
Internal standard	±1 Vdc for self-calibration; stability is ±0.006 percent per 6 months maximum at 25°C; temperature coefficient +0.001 percent/°C (+10° to +40°C). NOTE: 1 Vdc is the full scale output of the ac to dc voltage converter on all ranges. It is also a full scale output of the dc to digital converter (DVM).
Common mode noise rejection	110 dB at 60 Hz with 1,000Ω in common lead.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Dymec Division of Hewlett-Packard Model DY2401C (digital dc voltmeter) in conjunction with Model DY2410B (AC/ohms converter).

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ITEM W03

VOLTMETER, AC

(Obsolescent)

TYPE OF EQUIPMENT	AC digital voltmeter.
FUNCTION PERFORMED	Measurement of ac voltages (indicated rms voltage is proportional to absolute average of applied sine wave).
SPECIFICATIONS	
Voltage range	0.1 to 1,000 Vrms full scale in 5 ranges with 300 percent over-ranging without overload except on 1,000 V range which is usable to 750 v peak.
Frequency range	50 Hz to 100 kHz.
Accuracy	For +10° to 50°C (calibration at +25°C); 30 day stability (calibrated daily against internal standard); for 20 Hz: ±0.05 percent full scale ±0.35 percent reading; for 30 Hz: ±0.05 percent full scale ±0.25 percent reading; for 50 Hz to 10 kHz: ±0.05 percent full scale ±0.20 percent reading; for 10 to 30 kHz: ±0.06 percent full scale ±0.40 percent reading; for 30 to 100 kHz: ±0.10 percent full scale ±0.60 percent reading. NOTE: During over-ranging, full scale error is to be ignored and percent reading error is to be doubled.
Input impedance	1 MΩ shunted by 100 pF.
Internal standard	±1 Vdc for self-calibration; stability is ±0.01 percent per month (maximum) at 25°C; temperature coefficient ±0.002 percent/°C (+10° to +40°C). NOTE: 1 Vdc is the full scale output of the ac to dc voltage converter on all ranges. It is also a full scale output of the dc to digital converter (DVM).
Common mode rejection	110 dB at 60 Hz with 1,000Ω in common lead.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Dymec Division of Hewlett-Packard Model DY2401A (digital dc voltmeter) in conjunction with Model DY2410A (AC/ohms converter).

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ITEM W04

VOLTMETER, AC

TYPE OF EQUIPMENT	AC voltmeter.
FUNCTION PERFORMED	Broadband ac measurements (indicated rms voltage is proportional to absolute average value of applied sine wave).
SPECIFICATIONS	
Voltage range	1 mV to 300 V in 1, 3, 10 sequence (12 ranges).
Frequency range	10 Hz to 10 MHz.
Accuracy	Over temperature range from 0° to +55°C.
1 mV range at full scale	±1 percent from 40 Hz to 500 kHz; ±2 percent from 20 Hz to 1 MHz; ±4 percent from 2 Hz to 4 MHz; ±4 percent from 10 Hz to 10 MHz; -10 percent from 10 Hz to 10 MHz.
1 mV range at 1/3 full scale	±3 percent from 40 Hz to 200 kHz; $^{+3}_{-5}$ percent from 20 Hz to 200 MHz; $^{+3}_{-10}$ percent from 2 to 4 MHz; $^{+4}_{-10}$ percent from 10 to 20 Hz.
3 mV to 300 V ranges at full scale	±1 percent from 40 Hz to 1 MHz; ±2 percent from 20 Hz to 4 MHz; ±4 percent from 10 Hz to 10 MHz.
3 mV to 300 V ranges at 1/3 full scale	±3 percent from 40 Hz to 200 kHz; $^{+3}_{-5}$ percent from 20 Hz to 4 MHz; $^{+4}_{-10}$ percent from 10 Hz to 10 MHz.
Isolation	Instrument may be battery-operated if isolation from power lines is desired.
Input impedance	10 MΩ shunted by 21 pF on 1 mV to 1 V ranges; 10 MΩ shunted by 8 pF on 3 to 300 V ranges.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 400E* Model 400 EL.

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ITEM W05

VOLTMETER, AC

TYPE OF EQUIPMENT	AC voltmeter.
FUNCTION PERFORMED	Meter measures the rms value (heating value) of the input signal for all waveforms.
SPECIFICATIONS	
Voltage range	12 full scale ranges from 1 mV to 300 V in a 1, 3, 10 sequence.
Frequency range	10 Hz to 10 MHz.
Accuracy	±1 percent of full scale from 50 Hz to 1 MHz ±2 percent of full scale from 1 to 2 MHz ±3 percent of full scale from 2 to 3 MHz ±5 percent of full scale from 3 to 10 Hz ±5 percent of full scale from 3 to 10 MHz
Input impedance	10 MΩ shunted by 40 pF for 0.001 to 0.3 V range. 10 MΩ shunted by 40 pF for 1.0 to 300 V range.
Crest factor	(Ratio of peak amplitude to rms amplitude): 10:1 @ full scale except where limited by maximum input.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Hewlett-Packard Model 3400A.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	John Fluke Models 931A, 931B, 931P, and 931B rms ac-differential voltmeters.
Exceptions	Input voltage range: 0.01 to 1,100 Vrms, in five ranges of 0.1, 1, 10, 100, and 1,100 Vrms, each with 10 percent over-ranging. Frequency range is from 10 Hz to 1 MHz. Input impedance is 1 MΩ in parallel with 8 pF.

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ITEM W06

VOLTMETER, AC

(Obsolescent)

TYPE OF EQUIPMENT	AC voltmeter
FUNCTION PERFORMED	Measurement of ac voltage from 1 V to 1,000 V full scale.
SPECIFICATIONS	
Accuracy	±0.5 percent of reading.
Range	1 to 1,000 V.
Frequency bandpass	0 to 20 kHz.
Input impedance	1 MΩ.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Wavetek Model 210 with Model 2102 AC/DC/Ratio plug-in assembly.
Exceptions	Accuracy - ±0.2 percent of reading +25 uV at 20 Hz to 10 kHz. ±1 percent of reading +25 uV at 10 to 20 kHz. Input impedance 1 MΩ and 40 pF. Special features - Six-digit readout null balance instrument. This instrument can also measure dc voltage and dc ratios. With the proper plug-in unit frequency can also be measured.

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ITEM W07

VOLTMETER, TRUE RMS

TYPE OF EQUIPMENT - - - - - - - - - - - - - - AC voltmeter

FUNCTION PERFORMED

True rms - - - - - - - - - - - - - - - - - - Measures true rms using thermopile technique, 2 Hz to 100 MHz.

True rms of ac + dc - - - - - - - - - - - - Reading = $\sqrt{Vdc^2 + Vrms^2}$

dB, variable reference - - - - - - - - - - Standard calibration reads dBV (1 V = 0 dB). Capable of calibration to read: dBm 600Ω, dBm 75Ω, dBm 50Ω, and at least a 10 dB downward reference shift for comparison measurements. All controls front panel mounted.

Input impedance
Below 10 MHz - - - - - - - - - - - - - - (a) 1 to 1,000 V range - 10 MΩ ±10 percent shunted by 19 pF ±10 percent.
(b) 10 mV and 100 mV range - 20 MΩ ±10 percent shunted by 16 pF ±10 percent.

10 MHz to 100 MHz - - - - - - - - - - - - The following table shows maximum loading error due to input shunt impedance across a terminated source.

System impedance	Frequency	
Source and load	10 MHz	100 MHz
50	1%	10%
75	2%	20%

Crest factor (ratio of peak to rms) - - - (a) 2 to 25 Hz - 2:1 (full range input).
(b) Above 25 Hz - 10:1 (full range input).
(c) An overload circuit is provided to protect instrument input in the event that a and b above are exceeded.

Maximum input voltage
High to low - - - - - - - - - - - - - - - 1,000 Vrms, 1,500 v peak or 10^8 (V x Hz) on any range. Maximum dc in ac mode is ±500 Vdc.

Low to chassis - - - - - - - - - - - - - ±500 v peak.

Automatic ranging - - - - - - - - - - - - - Up range at approximately 190 percent of full range and down range at approximately 17 percent of full range.

Full range display - - - - - - - - - - - - 10.00 mV (ac only), 100.0 mV, 1.000 V, 10.00 V, 100.0 V, 1,000 V and -48 dBV to +60 dBV.

Accuracy
Volts - - - - - - - - - - - - - - - - - - - Reading accuracy = ±(percent of range ± percent of reading). 90 days (25° ±5°C). Less than 95% relative humidity. 17 to 190% of range. (See following table).

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W07 - Continued

Percent of range				Percent of reading						
Function				Frequency						
Range	DC	DC + AC	AC	DC	2Hz	25 Hz	100 kHz	1 MHz	10 MHz	100 MHz
10 mV	N/A	N/A	0.4	N/A	N/A	0.3	1.0	3.0	N/A	N/A
100 mV	0.6	0.6	0.4	0.2	0.4	0.2	0.5	2.0	2.0	10.0
1 V	0.2	0.2	0.2	0.2	0.4	0.2	0.5	1.0	2.0	10.0
10 V	0.2	0.2	0.2	0.2	0.4	0.2	0.5	1.0	N/A	N/A
100 V	0.2	0.2	0.2	0.2	0.4	0.2	1.0	N/A	N/A	N/A
1,000 V	0.3	0.3	0.3	0.2	0.4	0.2	N/A	N/A	N/A	N/A

dB - Reading accuracy = ±(0.15 dB + dB error vs frequency and range). (See table below).

	dB error versus frequency range						
Range	DC	2 Hz	25 Hz	100 kHz	1 MHz	10 MHz	100 MHz
10 mV	N/A	N/A	0.03	0.1	0.3	N/A	N/A
100 mV	0.02	0.04	0.02	0.05	0.2	0.2	1
1 V.	0.02	0.04	0.02	0.05	0.1	0.2	1
10 V	0.02	0.04	0.02	0.05	0.1	N/A	N/A
100 V	0.02	0.04	0.02	0.1	N/A	N/A	N/A
1,000 V	0.02	0.04	0.02	N/A	N/A	N/A	N/A

Outputs
Display - - - - - - - - - - - - - - - - - Digital, 3 digit with 4th for over-range.
Recorder out - - - - - - - - - - - - - - 1 Vdc = Full range.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard Model 3403C with options 001 and 006.

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

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ITEM X01

AMMETER, DC

TYPE OF EQUIPMENT - - - - - - - - - - - - - Direct current microammeter.

FUNCTION PERFORMED - - - - - - - - - - - - Accurately measures current over a wide range.

SPECIFICATIONS

Current range - - - - - - - - - - - - - - - +1 uA to 1.0 A full scale in 13 ranges, (2 steps/decade; 1, 3 sequence).

Accuracy - - - - - - - - - - - - - - - - - - ±2 percent full scale on any range.

Input resistance - - - - - - - - - - - -

Range	Internal shunt resistance*
0.001 mA	1,000Ω
0.003 mA	316Ω
0.01 mA	100Ω
0.03 mA	31.6Ω
0.1 mA	10Ω
0.3 mA	3.16Ω
1 mA	1Ω
3 mA	0.316Ω
10 mA	0.1Ω
30 mA	0.1Ω
100 mA	0.1Ω
300 mA	0.1Ω
1,000 mA	0.1Ω

* For total internal resistance, add 0.07Ω copper lead resistance at 25°C.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Hewlett-Packard Model 412A (HP412 obsolescent).

EQUIPMENT MEETING SPECIFICATIONS
WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - - Conu Model 208R.
Exceptions - - - - - - - - - - - - - - - - Ammeter: Full scale current range = 100 pA to 100 mA (19 ranges).
Input resistance = 10 kΩ, 100 pA to 10 uA ranges.
Input resistance = 10Ω, 30 uA to 100 mA ranges.

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ITEM X02

AMMETER, DC

TYPE OF EQUIPMENT	Direct current ammeter.
FUNCTION PERFORMED	The meter presents a digital display of the measured current.

SPECIFICATIONS

Current Range	1.0000 mA full scale +20 percent over-range 10.000 mA full scale +20 percent over-range 100.00 mA full scale +20 percent over-range 1,000.0 mA full scale.
Accuracy	±0.13 percent of reading ±0.02 percent full scale.
Resolution	0.1 uA.
Temperature coefficient	±0.003 percent of reading ±0.0008 percent full scale/°C.
Input resistance	1 kΩ @ 1 mA scale 100Ω @ 10 mA scale 10Ω @ 100 mA scale 1Ω @ 1,000 mA scale
Overload protection	Maximum of 2A may be safely applied to any range without damage.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer	Fairchild Model 7000 Digital multimeter with option 04 plug-in board.

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ITEM X03

AMMETER, DC

TYPE OF EQUIPMENT	DC microammeter.
FUNCTION PERFORMED	Measurement of low level dc current to ±5 percent of reading.
SPECIFICATIONS	
Accuracy	±5 percent of reading or better.
Range	10^{-10} A full scale to 10^{-3} A.
Input resistance	Less than 1Ω.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Keithley Instruments Model 150A microvoltammeter. Accuracy ±3 percent of full scale (worst case accuracy occurs at 3/10 of full scale. Therefore, on a percent reading basis, the instrument accuracy is ±10 percent of reading or better.) Input resistance 1Ω on 10^{-3} A range 10^{6}Ω on 10^{-10} A range. Special features - This instrument can also be used as follows: DC voltmeter (Item V05) DC amplifier Null detector

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ITEM X04

AMMETER, DC

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Direct current ammeter.

FUNCTION PERFORMED - - - - - - - - - - - - - Accurately measures current over a wide range.

SPECIFICATIONS

Current range - - - - - - - - - - - - - - - - ±75 uA to 15.0 A fs in 12 ranges, (2 steps/ decade; 1.5, 7.5 sequence).

Accuracy - - - - - - - - - - - - - - - - - - ±2 percent fs on any range.

Input resistance - - - - - - - - - - - - - -

Range	Internal shunt resistance
75.00 uA	2.000 kΩ
150.00 uA	2.000 kΩ
.75 mA	200.000Ω
1.50 mA	200.000Ω
7.50 mA	21.300Ω
15.00 mA	21.300Ω
75.00 mA	2.140Ω
150.00 mA	2.140Ω
.75 A	.214Ω
1.50 A	.214Ω
7.50 A	.021Ω
15.00 A	.021Ω

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Simpson Model 263.

EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer - - - - - - - - - - - - - - - Simpson Model 270.
Exceptions - - - - - - - - - - - - - - - - Full scale current range of 50 uA to 10 A in six ranges.

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ITEM X05

AMMETER, DC

TYPE OF EQUIPMENT	DC ammeter, digital.
FUNCTION PERFORMED	High current measurement capability.
SPECIFICATIONS	
Current ranges	± 199.9 uA ± 1.999 mA ± 19.99 mA ± 199.9 mA ± 1999 mA ± 10.00 A (1 min operation from 10 to 20 A).
Accuracy (1 year 15° to 35°C)	± (0.3 percent of reading + 1 digit) all ranges except 10 A. ± (0.5 percent of reading + 1 digit) on 10 A range.
Voltage burden (over voltage drop at terminals	0.5 V maximum up to 10 A.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	John Fluke Model 8000A-05. John Fluke Model 8000A-015.

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ITEM Y01

PROBE, WIDEBAND

TYPE OF EQUIPMENT - - - - - - - - - - - - - - Wideband voltage probe.

FUNCTION PERFORMED - - - - - - - - - - - - - To enable voltage monitoring with minimum circuit loading and signal distortion.

SPECIFICATIONS

Input impedance - - - - - - - - - - - - - - 10 MΩ (minimum) in shunt with 6 pF (maximum).

Attenuation - - - - - - - - - - - - - - - - 1X

Bandwidth - - - - - - - - - - - - - - - - - DC to 230 MHz (minimum).

Flatness - - - - - - - - - - - - - - - - - - ±0.4 dB across the band.

Maximum input voltage - - - - - - - - - - - ±100 v peak (minimum).

Noise - - - - - - - - - - - - - - - - - - - Less than 1.5 mV.

Offset - - - - - - - - - - - - - - - - - - - Adjustable to ±1 Vdc.

Drift - - - - - - - - - - - - - - - - - - - Less than 0.5 mV/°C.

Signal delay time - - - - - - - - - - - - - 12 ns (maximum) from probe tip to amplifier input.

EQUIPMENT MEETING
ALL SPECIFICATIONS

Manufacturer - - - - - - - - - - - - - - - Tektronix Type P6045.
Input impedance - - - - - - - - - - - - - - 10 MΩ shunted by

Attenuation	DC	AC	Maximum input
X1	4 pF	6 pF	±2 V
X10	2 pF	4 pF	±20 V
X100	1.6 pF	3.6 pF	±100 V

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ITEM T02

PROBE, OSCILLOSCOPE VOLTAGE

TYPE OF EQUIPMENT	General purpose 10X voltage probe.
FUNCTION PERFORMED	To enable voltage monitoring with minimum circuit loading and signal distortion.
SPECIFICATIONS	
Input impedance	10 MΩ in shunt with an adjustable capacitor.
Attenuation	10X ±2 percent.
Bandwidth	DC to 70 MHz.
Maximum input voltage	±600 v peak.
Noise	<1.5 mV.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Tektronix Type P6006 Tektronix Type P6008 Tektronix Type P6010 Tektronix Type P6012 Tektronix Type P6032 Tektronix Type P6047.
Manufacturer	Hewlett-Packard Type 10001A Hewlett-Packard Type 10003A Hewlett-Packard Type 10004A.

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ITEM Y03

PROBE, OSCILLOSCOPE CURRENT

TYPE OF EQUIPMENT	General purpose 1 mA/div current probe.
FUNCTION PERFORMED	To enable current monitoring with minimum circuit loading and signal distortion.
SPECIFICATIONS	
Sensitivity	Adjustable from 1 mA/div to 1A/div with 50 mV/div oscilloscope sensitivity.
Accuracy	±3 percent.
Bandwidth	DC to 50 MHz.
Maximum input current	±10 A peak.
Maximum circuit voltage	600 V peak.
Noise	<0.5 mA.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Tektronix Type P6042.

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ITEM Y04

PROBE, OSCILLOSCOPE VOLTAGE

TYPE OF EQUIPMENT	General purpose 1X voltage probe.
FUNCTION PERFORMED	To enable voltage monitoring with minimum loading and signal distortion.
SPECIFICATIONS	
Input impedance	1 MΩ in shunt with 50 pF maximum.
Attenuation	1X.
Bandwidth	DC to 28 MHz.
Maximum input voltage	±600 v peak.
Noise	<1.5 mV.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Tektronix Type P6011 Tektronix Type P6027 Tektronix Type P6028 Tektronix Type P6045 - Item Y01.

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ITEM Y05

PROBE, OSCILLOSCOPE CURRENT

TYPE OF EQUIPMENT	Wide band current probe.
FUNCTION PERFORMED	To enable current monitoring with minimum circuit loading and signal distortion.
SPECIFICATIONS	
Sensitivity	1 mV/mA minimum.
Accuracy	±3 percent.
Bandwidth	10 Hz to 200 MHz.
Maximum input current	±5 a peak.
Maximum circuit voltage	600 v peak.
Noise	<0.5 mA.
EQUIPMENT MEETING ALL SPECIFICATIONS	None.
EQUIPMENT MEETING ALL SPECIFICATION WITH EXCEPTIONS AS NOTED	
Manufacturer	Tektronix Type P6021.
Exception	Bandwidth 120 Hz to 60 MHz.
Manufacturer	Tektronix Type P6022.
Exception	Bandwidth 935 Hz to 200 MHz.
Manufacturer	Tektronix Type P6041
Exception	Bandwidth 1.2 kHz to 200 MHz.
Manufacturer	Hewlett-Packard Model 456A.
Exception	Bandwidth 25 Hz to 20 MHz.
Manufacturer	Hewlett-Packard Model 1110A with Model 1111A.
Exception	Bandwidth 50 Hz to 20 MHz.

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ITEM Z01

VOLTMETER, PHASE ANGLE

TYPE OF EQUIPMENT	Phase angle voltmeter [PAVM].
FUNCTION PERFORMED	For the measurement of complex ac voltages, phase angles, and nulls in the phase sensitive mode, and for measurement of broadband ac voltages in the total mode.

SPECIFICATIONS

Voltage range	1 mV full scale to 300 V full scale in 12 ranges.
Frequency range	
Total mode	10 Hz to 50 kHz direct (useful frequency range with input transformer 50 Hz to 10 kHz).
Phase sensitive mode	400, 800, 10,000 Hz with ±5 percent bandwidth.*
	* Unlike Z02 which needs no adjustment over a 5 percent bandwidth in order to meet all Z01 specifications, the item Z01 PAVM must be trimmed at any frequency encompassed by the specified bandwidth to meet all specifications.
Accuracy	
Voltage	
Total mode	±2 percent of full scale, direct (10 Hz to 50 kHz).
Fundamental	±2 percent at frequency of operation.
Phase sensitive mode	±1° as read on the phase dial; ±3 percent of full scale angle when using E Cos θ characteristic.
Calibration	
Voltage	Zero center, low friction meter calibrated with 3-0-3 and 10-0-10 scales.
Phase dial	Continuously calibrated from -6° to +96° over four quadrants (0°, 90°, 180°, and 270°) with 1° divisions.
Signal input impedance	
Direct mode	10 MΩ at resonance, nominal.
Transformer mode	300 kΩ at 400 Hz, 1 MΩ at resonance, nominal.
Reference input impedance	100 kΩ ±10 percent (see input voltage levels, transformer mode).
Reference input voltage range	(With reference isolation tranformer) minimum voltage is 1.25 Vrms, maximum voltage is 2.0 x F or 150 Vrms, whichever is less.
Isolation	Signal and reference isolation (except when in direct input mode, then only reference is isolated).
Input voltage levels	
DC volts	
Direct mode	400 Vdc maximum.
Transformer mode	0 V.

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AC volts	
Direct mode	300 Vrms maximum.
Transformer mode	1.0 x F or 300 V, whichever is less. All dc components must be blocked.
Overload	Overload lamp set at approximately 5.5 x full scale voltage for all ranges.
Noise	Less than 5 µV for low-impedance inputs.
Nulling sensitivity	5 µV (2 µV in phase sensitive mode).
Harmonic rejection	55 dB or greater.
Orthogonality	0.1°.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	North Atlantic Model *VM204 S462 Model 2146 S1281 Model VM204 S481.

EQUIPMENT MEETING SPECIFICATION
WITH EXCEPTIONS AS NOTED

Manufacturer	North Atlantic.
Exceptions	Most North Atlantic voltmeters in the VM202, VM204, VM214, and VM301 series can meet the phase angle voltmeter requirements above at one or more of the specified frequencies (400, 800, and 10 kHZ) provided they are equipped with the following: (a) Matched low-pass filters in both the signal and reference channels for one or more of the specified frequencies. (b) Reference isolation transformer. (c) Programmable input isolation transformer. The required accessories need not be integral to the voltmeter. However, the selected voltmeter utilizing the accessories shall meet the above listed specifications for the frequency or frequencies at which the instrument is used.

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ITEM Z02

VOLTMETER, PHASE ANGLE

TYPE OF EQUIPMENT	Phase angle voltmeter (PAVM).
FUNCTION PERFORMED	For the measurement of complex ac voltages, phase angles, and nulls in the phase sensitivity mode, and for measurement of broadband ac voltage in the total mode.
SPECIFICATIONS	
Voltage range	300 μV full scale to 300 V full scale in 13 ranges.
Frequency range	
Total mode	10 Hz to 100 kHz direct (see signal input impedance, isolated, for input transformer characteristics).
Phase sensitive mode	400, 800, and 9,700 Hz with ±5 percent bandwidth.
Accuracy	
Voltage	
Total mode	±2 percent of full scale, direct 10 Hz to 100 kHz).
Fundamental	±2 percent at frequency of operation.
Phase sensitive mode	±0.5° as read on the phase dial at 400 and 800 Hz; ±1° as read on the phase dial at 9.7 kHz; ±3 percent of full scale angle when using E Cos θ characteristic.
Calibration	
Voltage	Zero center, low friction meter calibrated with 3-0-3 and 10-0-10 scales.
Phase dial	Continuously calibrated from -6° to 96° over four quadrants (0°, 90°, 180° and 270°) with 0.5° divisions.
Signal input impedance	
Direct mode	10 MΩ shunted by 75 pF, nominal.
Isolated	350 kΩ at 400 Hz (30 Hz to 2.5 kHz). 50 kΩ at 4.8 kHz (2.5 to 20 kHz).
Transformer mode	
Low frequency range (30 to 2,500 Hz)	100 kΩ at 400 Hz in the 10 to 300 V ranges; 500 kΩ at 400 Hz in all other ranges.
High frequency range (30 Hz to 20 kHz)	(See input voltage levels, transformer mode) 250 kΩ at 2,500 Hz.
Reference input impedance	100 kΩ shunted by 120 pF nominal.
Reference input voltage range	1 to 150 Vrms maximum.
Isolation	Signal and reference are isolated except when in direct input mode, then only reference is isolated.
Input voltage levels	
DC volts	
Direct mode	400 Vdc maximum.
Transformer mode	0 V.

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AC volts	
Direct mode	300 Vrms maximum.
Transformer mode	1.0 x F or 300 V, whichever is less. All dc components must be blocked.
Overload	10 x full scale voltage in any range (indicator lights at approximately 11 x full scale volts).
Noise	Less than 15 μV total and fundamental modes.
Nulling sensitivity	Less than 2 μV in the phase sensitive mode.
Harmonic rejection	55 dB or greater.
Orthogonality	0.1°.

EQUIPMENT MEETING ALL SPECIFICATIONS

Manufacturer	North Atlantic Model 214C-S1281.

EQUIPMENT MEETING ALL SPECIFICATIONS WITH EXCEPTIONS AS NOTED

Manufacturer	North Atlantic
Exceptions	The VM301 voltmeter can meet all but one of the above listed specifications at one or more of the specified frequencies (400, 800, and 10,000 Hz) provided it is equipped with the following: (a) Matched low-pass filters in both the signal and the reference channels for one or more of the specified frequencies. (b) Reference isolation transformer. (c) Programmable isolation transformer. The required accessories need not be integral to the voltmeter. However, the selected voltmeter utilizing the accessories shall meet the above listed specifications for the frequency, or frequencies, at which the instrument is used. One exception generally applicable is to the 300 μV full scale sensitivity. The VM301's lowest full scale voltage is 1 mV.

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ITEM Z03

PHASEMETER

TYPE OF EQUIPMENT	Phasemeter.
FUNCTION PERFORMED	Measures the difference in phase between two ac input signals.
SPECIFICATIONS	
Input voltage range	0.01 to 300 Vrms with frequency of 5 Hz to 10 kHz. 0.01 to 50 Vrms with frequency of 10 to 500 kHz.
Frequency range	5 Hz to 500 kHz.
Angle ranges	0.00° to ±180.00° or 0.00° to +360° automatically selected.
Angle resolution	0.01 degrees.
Phase measurement accuracy Both input signal levels equal	±0.1° with input levels from 0.05 to 50 Vrms and frequency of 50 Hz to 50 kHz. ±0.1° with input levels from 0.01 to 0.05 Vrms or 50 to 300 Vrms and/or frequency of 5 to 50 Hz or 50 to 500 kHz.
Input impedance	1 MΩ shunted with 40 pF.
Error rejection	False triggers due to high-frequency noise are locked out for signal to noise ratios better than 10:1.
Harmonic rejection	Error less than 0.06°/% distortion for 2nd harmonic. Error less than 0.13°/% distortion for 4th harmonic. Error less than 0.2°/% distortion for 6th harmonic. Odd harmonics in phase with fundamental cause no error.
EQUIPMENT MEETING ALL SPECIFICATIONS	
Manufacturer	Dranetz Model 305 with 305-PA-3001 plug-in.
EQUIPMENT MEETING SPECIFICATIONS WITH EXCEPTIONS AS NOTED	
Manufacturer	Krohn-Hite Model 6500.
Exceptions	Input voltage range: 0.1 to 120 Vrms. Frequency range: 10 Hz to 5 MHz. Angle ranges: 0.00° to 360.00°. Phase measurement accuracy: ±0.05° with frequency from 20 Hz to 50 kHz. ±0.20° with frequency from 10 to 20 Hz and from 50 to 100 kHz. Beyond 100 kHz, accuracy drops at the rate of ±0.7°/100 kHz.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 AR 1426

PREVIOUS EDITION IS OBSOLETE.

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